

Monetary Policy and the Credit Rationing Effects of Liquidity

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Abstract

This paper studies how liquidity provision affects credit rationing and monetary policy in a New Keynesian economy with frictional bank lending. Liquidity provision lowers banks' funding costs but can also compress the spread between the market rate and interest on reserves, worsening adverse selection and credit rationing. The resulting trade-off can make quantitative easing contractionary away from the effective lower bound, while easing at the bound stimulates activity through inflation expectations. These distortions shape optimal policy. Under discretion, restraining liquidity provision to preserve lending margins generates a deflation bias. Under commitment, forward guidance eases current credit conditions by raising expected lending profitability, and the obligation to deliver on past commitments implies gradualism in quantitative tightening.

JEL: E5, E44, G21

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1 Introduction

The monetary policy environment has evolved considerably since the global financial crisis (GFC). Many central banks expanded their menu of policies and now routinely use balance sheet policies to respond to crises. Yet the effects of these policies remain imperfectly understood, with implications both for the design of future interventions and for the unwinding of central bank balance sheets. While these policies have led to a dramatic rise in central bank balance sheets and banking sector liquidity, a central challenge has been ensuring that these expansions translate into improved lending conditions, particularly for small businesses. In practice, despite a sharp increase in banking sector liquidity after the GFC, euro area banks remained reluctant to lend, even after the easing of the sovereign debt crisis. Figure 1 illustrates this disconnect, showing weaker lending (panel a) alongside a substantial and sustained rise in excess reserve balances (panel b).

This is difficult to reconcile with most macroeconomic theories of bank lending, which predict that liquidity expansions drive more lending and investment (see, e.g., Kashyap et al., 2002;

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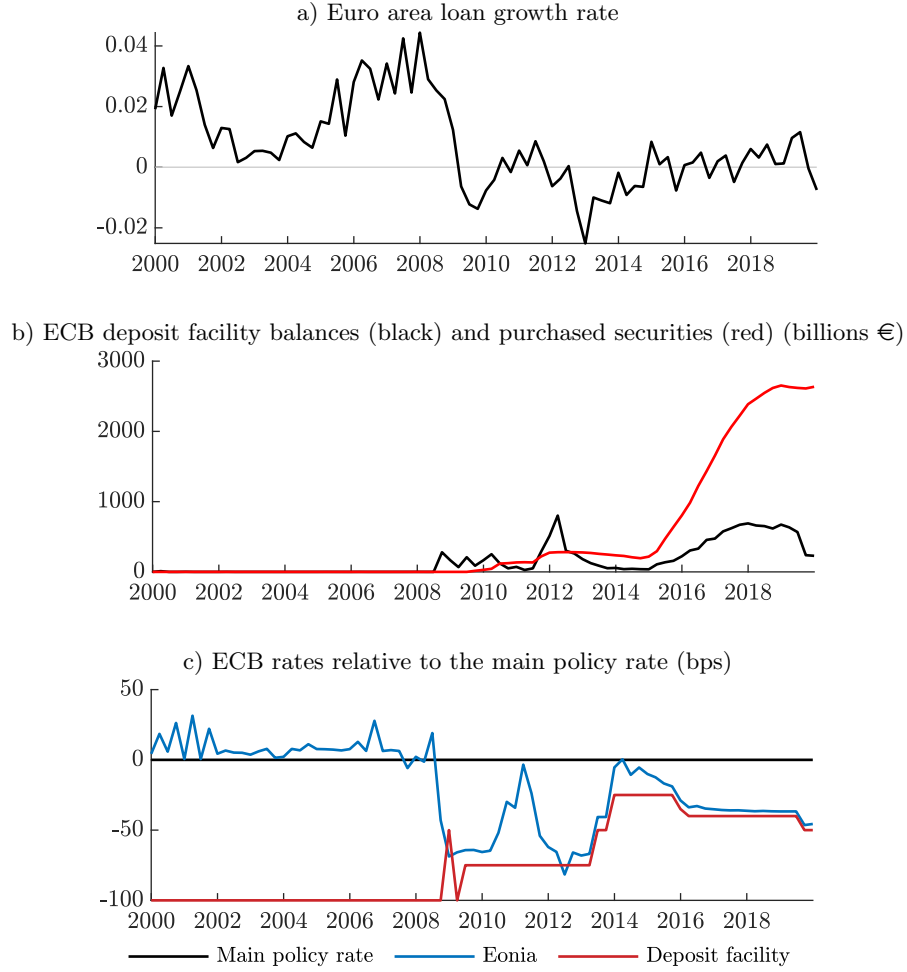


Figure 1: Source: ECB. Loans are those to euro area NFCs reported by MFIs. Purchased securities includes the stock of securities purchased as part of the various asset purchase programmes since 2009.

Adrian and Shin, 2010; Carpenter et al., 2014), yet recent evidence suggests this is not always the case. For example, Diamond et al. (2024) find that bank reserve injections crowded out bank loans in the US, while Iyer et al. (2014) show that greater euro area liquidity from ECB balance sheet policies led to liquidity hoarding rather than increased small-business lending.¹

In this paper, we rationalise this evidence using a New Keynesian macroeconomic model in which information frictions in small business lending interact with monetary policy implementation. The key mechanism is that policy-induced changes in bank profitability alter the opportunity cost of lending, generating endogenous credit rationing even in the presence of abundant liquidity.² The lending friction prevents banks from observing the risk characteristics of small firms in the economy, creating an adverse selection problem in credit markets. This generates a well-established effect, explored by Ikeda (2020) and Swarbrick (2023) in relation to the GFC, under which banks ration credit and build up excess reserves when firm risk is high. The key

¹Similarly, Acharya et al. (2019) find that the additional liquidity led to higher cash reserves and banks holding low-interest assets, but did not lead to more productive business lending. See also Gambacorta and Marques-Ibanez (2011), Joyce et al. (2011), Loutskina (2011), and Disyatat (2011). Mennuni (2025) develops a theory rationalising the buildup of excess bank liquidity.

²Evidence that adverse selection has been an important feature of European credit markets motivates this focus (see Albertazzi et al., 2021).

determinant in this model is the spread to the floor: the gap between the market rate and the rate paid on reserves, which equals the policy corridor absent liquidity operations. Because reserves offer a risk-free outside option, this spread determines the opportunity cost of extending risky loans, and therefore the incentive to ration credit to small businesses.

An important implication is that a narrowing of the interest corridor, such as when policy rates approached the effective lower bound (ELB) (Figure 1, panel c), can increase the incentive to ration small business loans, offsetting the expansionary effects of low rates and the unconventional policy measures. This creates a tension whereby when the bank funding rate falls, activity may be stimulated through the standard interest rate channel, but lending incentives may simultaneously weaken as rates approach the floor, compressing net margins and increasing the relative attractiveness of holding reserves. This mechanism is supported by empirical evidence presented in Section 2, which shows that while central bank asset purchases and falling bank funding costs predict more lending in the euro area, a compression of the interest rate spread to the floor is a significant predictor of the opposite outcome.

Using a restricted version of the model, we show that a log-linear approximation takes the form of a 3+1 equation New Keynesian model, where the standard IS and Phillips curves are augmented by a credit wedge condition that captures endogenous lending distortions.³ Because the credit wedge acts as a one-sided cost-push distortion, it generates a trade-off between inflation stabilisation and output stabilisation that is absent when lending is unconstrained. This trade-off has implications for optimal monetary policy design: under discretion, a policymaker facing credit rationing at the ELB optimally restrains liquidity provision to preserve bank lending margins, generating a deflation bias whose severity depends on the degree of adverse selection. Under commitment, forward guidance can ease credit conditions directly by raising expected lending profitability, and the obligation to deliver on past commitments provides a rationale for gradualism in the subsequent unwinding of balance sheet positions.

This paper makes three main contributions to our understanding of monetary policy transmission in the presence of financial frictions. First, we provide motivating evidence that the spread between the market rate and the policy floor predicts lending in the euro area, so that a market rate falling towards the floor predicts less lending rather than more. Second, we embed the adverse-selection banking friction of Swarbrick (2023) in a New Keynesian model in which reserve remuneration and the spread to the floor shape the opportunity cost of lending, opening a credit-rationing channel that can make liquidity provision contractionary. Third, we show how this interaction changes optimal policy design: easing is moderated by the incentive to ration credit.

These contributions rest on a framework that departs from existing models in several respects. Much of the large literature examining the links between liquidity and macroeconomic outcomes focuses on how private provision of liquidity can tighten during crises, amplifying shocks and motivating a stabilising role for liquidity easing policies (see, e.g. Bernanke and Gertler, 1989; Kiyotaki and Moore, 1997; Holmström and Tirole, 1998; Adrian and Shin, 2010; Gertler and

³This shares structural similarities with Sims et al. (2023), who derive a four-equation New Keynesian model in which QE and credit shocks enter both the IS and Phillips curves.

Karadi, 2011; Brunnermeier and Sannikov, 2014). Recent work has also emphasised the macroeconomic importance of adverse selection in external finance, which can distort the allocation of credit and investment (see e.g. Guo et al., 2025; Beyhaghi et al., 2026). The model developed in this paper builds on Stiglitz and Weiss’s (1981) work on adverse selection, which shows how credit markets can shut down due to information frictions.⁴

In contrast to models that link monetary policy transmission to borrower balance sheets through a financial accelerator, whereby monetary expansions raise asset prices and collateral values (e.g., Bernanke et al., 1999; Christiano et al., 2005), this paper abstracts from the balance sheet channel. Here, the effects of a monetary expansion are ambiguous as they depend more critically on how the expansion is implemented and how it affects banking sector liquidity, bank balance sheet composition, bank profitability, and the opportunity cost of lending. The key issue is how the shock or policy action interacts with banks’ incentive to ration credit.

The theoretical model incorporates monetary policy implementation in the presence of information frictions. Banking sector liquidity is endogenous to macro-financial conditions, and banks may choose to hold excess reserves in equilibrium, so the interest rate paid on reserves affects bank profitability, liquidity demand, and ultimately lending behaviour (see also Disyatat, 2011). Standard New Keynesian models abstract from these details and contain no endogenous excess reserves, while existing work on the interest rate corridor (see, e.g., Bindseil, 2000; Whitesell, 2006) focuses on trade-offs between interest rate volatility, interbank market activity, and monetary transmission under frictions such as search costs (Bech and Monnet, 2016) and balance sheet risk (Whitesell, 2006; Bindseil and Jablecki, 2011). In contrast, this paper shows how the corridor affects lending conditions through a profitability channel, even in the absence of interbank frictions: the floor rate competes with risky lending as an alternative use of liquidity, and this trade-off shapes banks’ willingness to intermediate.

The profitability channel connects the analysis to work on the contractionary effects of low policy rates (e.g., Abadi et al., 2023; Eggertsson et al., 2024), where a binding floor on deposit rates compresses margins as the policy rate falls. In contrast, the mechanism here is triggered by the spread to the floor rather than the level of the policy rate, and this spread compresses whenever liquidity is abundant. The contractionary region is therefore away from the effective lower bound, with liquidity provision remaining expansionary at the bound. A similar distinction applies to the crowding out of lending by reserve injections, which Diamond et al. (2024) attribute to balance sheet costs, whereas here it operates through the opportunity cost of lending. Section 2 examines bank-level evidence relevant to these competing explanations.

2 Motivating evidence from the euro area

The motivating facts are shown in Figure 1: reserve balances rose sharply and persistently, the overnight rate fell to the policy floor, and business lending remained weak. In this section we document a systematic relationship between the spread to the floor and banks’ lending and

⁴This has been extended in many papers in the years since, including, for example Bester (1985), Mankiw (1986), Williamson (1986), De Meza and Webb (1987), Besanko and Thakor (1987), House (2006), Ikeda (2020).

reserve holdings in the euro area, controlling for macroeconomic conditions, credit risk, and ECB asset purchases. We then ask whether that relationship can be attributed to the mechanism proposed here rather than to competing explanations.

Focusing on the euro area is insightful due to the important role of banks in providing credit to the real economy. Around double the share of business debt takes the form of bank loans compared to the United States,⁵ while small and medium-sized enterprises account for around two thirds of private-sector employment in Europe against around one half in the United States (see Carbo-Valverde and Rodríguez-Fernández, 2016).

The euro area is also a useful laboratory because the ECB’s operating framework has generated rich variation in the return on reserves. The ECB operates a corridor system, and during periods of abundant liquidity the overnight interbank market rate falls below the main policy rate, so that the spread between the interbank rate and the policy floor becomes a key determinant of banks’ opportunity cost of lending. As a result, changes in the floor rate and the effective spread to the floor can affect lending incentives even when conventional policy rates move little (see Figure 1, panel c).

2.1 The Interbank Rate and the Policy Floor

To explore the relationship between the spread to the floor and lending conditions, we estimate the following specification using quarterly bank-level data between 2008 and 2019:

$$\ln X_{i,t} = \alpha_{X,i} + \beta_X (R_t - \underline{R}_t) + \delta_X R_t + \gamma'_X \cdot \mathbf{z}_t + u_{X,i,t}, \quad (2.1)$$

for $X \in \{L, Res\}$, where $L_{i,t}$ is total gross loans and $Res_{i,t}$ is cash and balances at the central bank.⁶ R_t is the interbank market rate (Eonia) and $\underline{R}_t$ the interest on reserves.⁷ $\alpha_{X,i}$ denotes bank fixed effects, and $\mathbf{z}$ is a vector of controls including GDP, inflation, credit risk, and ECB asset purchases.⁸

Table 1 reports the results. Conditional on the interbank rate, a 25 basis point wider spread predicts around 4.8% higher lending and much lower reserve balances.⁹ Because the specification conditions on the interbank rate, the coefficient on the spread isolates movements in the policy floor, the rate paid on reserves. The positive lending coefficient is therefore consistent with the opportunity-cost channel: lending is higher when the return on holding reserves is lower relative to banks’ funding costs, even after controlling for the stock of Eurosystem asset purchases.

The loan column carries two coefficients of opposite sign. Conditional on the spread, a higher interbank rate predicts lower lending, whereas a wider spread predicts higher lending for a given

⁵70% vs. 35% (see Holm-Hadulla, Musso, Nicoletti and Tujula, 2022; Gambacorta, Yang and Tsatsaronis, 2014).

⁶Bank-level balance-sheet data are from S&P Global Market Intelligence (SNL Financial Fundamentals), covering banks in Austria, Belgium, France, Germany, Ireland, Italy, Netherlands, Portugal, Slovenia and Spain. See Appendix D.1 for normalisation, weighting and estimation details.

⁷That is, the interest rate on the ECB’s standing deposit facility.

⁸Controls are euro-area GDP, inflation, Gilchrist and Mojon’s (2018) measure of corporate credit risk, and the stock of Eurosystem asset purchases; see Appendix D.1 for definitions and sources.

⁹The loan coefficient is robust to a Newey-West correction for serial correlation. Further details in Appendix D.2.

	$\ln Res_t$	$\ln L_t$
$s_t \equiv R_t - \underline{R}_t$	-0.172^{***} (0.062)	0.048^{***} (0.013)
R_t	0.001 (0.021)	-0.019^{***} (0.004)
Obs.	5762	4161
Adj. R^2	0.919	0.291

Notes: The spread to the floor is the interbank rate, Eonia, minus the deposit facility rate, measured per 25 basis points. Balance-sheet variables are cash and balances with central banks, $\ln Res_t$, and total loans, $\ln L_t$, from SNL Financial Fundamentals. Both columns include bank fixed effects and the controls described in the text. Standard errors, in parentheses, are clustered by bank and by quarter. Significance levels: $***p < 0.01$, $**p < 0.05$, $*p < 0.1$.

Table 1: Estimation results: the spread to the floor and bank balance sheets

interbank market rate. The two have opposing implications whenever the interbank rate moves while the floor is held fixed. Combining the two, holding the floor fixed, a 25 basis point lower interbank rate predicts around 2.9% lower lending.¹⁰ A lower market rate predicting less lending is the reverse of the relationship a policy easing would usually imply, and it is this comovement that motivates the mechanism developed below.

We label these the funding-cost channel and the opportunity-cost channel. Our mechanism, developed in Section 3, focuses on the latter: because the interest on reserves is the return on banks' alternative to lending, a narrowing of the spread raises the opportunity cost of lending and can discourage it. This channel can rationalise the empirical findings, but the reduced-form evidence cannot distinguish it from other mechanisms that could generate the same observations.

2.2 Competing explanations and cross-sectional evidence

The comovement in Table 1 is consistent with the mechanism developed in this paper but does not by itself identify it. Following the GFC, the interbank rate fell towards the floor over a period of weak loan demand, interbank fragmentation, and strained bank balance sheets, and one in which the ECB eased in response to those conditions. The market rate compressing towards the floor, abundant reserves, and weak lending may therefore be joint symptoms of that environment rather than evidence that a narrowing spread reduced credit supply, since these conditions are only partially absorbed by the aggregate controls.

Reversal-rate and sticky-deposit models provide a further explanation, under which a lower market rate reduces lending by compressing banks' margins against a floored deposit rate rather than by raising the return on reserves (Drechsler et al., 2017; Abadi et al., 2023; Eggertsson et al., 2024).

The reserve reallocation points towards the opportunity-cost channel. With ample reserves, the return on the outside option to lending is the rate on reserves, so a spread compressing towards the floor raises the cost of lending relative to holding reserves, and the shift into central-bank

¹⁰The overall effect is the sum of the two coefficients, +0.029 ($p = 0.03$); see Appendix D.2.

	Bank + quarter FE (1)	Core/periph. × quarter FE (2)
$s_t \times$ reserve-richness	0.018** (0.007)	0.022*** (0.008)
$s_t \times$ deposit funding	0.003 (0.007)	0.002 (0.008)
$s_t \times$ capitalisation	0.003 (0.007)	-0.004 (0.007)
Banks	62	62
Obs.	1,784	1,784

Notes: Top quartile of the asset distribution. The dependent variable is the log loan share, $\ln(L_{i,t}/A_{i,t})$. Characteristics are standardised and bank-constant. All columns include bank fixed effects; column 1 adds quarter fixed effects, column 2 core/periphery-by-quarter. Standard errors clustered by bank in parentheses. The periphery is Ireland, Italy, Portugal, Spain and Slovenia. The core is Austria, Belgium, France, Germany and the Netherlands. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 2: Cross-sectional heterogeneity in the loan share

balances is the reallocation our mechanism predicts. While this result supports the proposed mechanism, it does not rule out the sticky-deposit channel. To assess whether the pattern of lending across banks supports the opportunity-cost channel rather than competing explanations, we exploit cross-sectional heterogeneity in banks' exposure to the spread, using time fixed effects to absorb the common aggregate environment.

We interact the aggregate spread with time-invariant bank characteristics: reserve-richness, which identifies the banks for which the return on reserves is the binding outside option, where our mechanism should operate most strongly, and deposit funding, which captures the sticky-deposit margin. A third interaction, with bank capitalisation, evaluates the role of bank stress, and of balance-sheet costs under which reserves crowd out lending in banks nearer their capital constraints. We estimate the specification:

$$\ln(L_{i,t}/A_{i,t}) = \alpha_i + \delta_t + \sum_k \theta_k (s_t \times Z_{k,i}) + u_{i,t}, \quad (2.2)$$

where $L_{i,t}/A_{i,t}$ is the loan share of assets, the natural margin for a reallocation between loans and reserves; α_i and δ_t are bank and quarter fixed effects; s_t is the spread to the floor; and the $Z_{k,i}$ are the standardised characteristics.¹¹ The bank effects absorb each characteristic's level and the quarter effects absorb all aggregate controls and anything else common across banks in a quarter, including the spread itself. θ_k is therefore identified only from the cross-section, measuring how banks of different types respond to the same spread. A positive θ_k marks a type that moves out of loans by more as the spread to the floor compresses.

Table 2 reports the interactions for the larger banks, the top quartile by assets, where the lending margin is most active.¹² For each 25 basis point compression of the spread, a bank one

¹¹Each characteristic is bank-constant, measured as bank i 's average ratio over 2008–2010 and standardised across banks; see Appendix D.1 for definitions.

¹²The top quartile captures around 94% of total assets in the sample. The reserve interaction is positive across the size distribution and significant through the upper quartile; the full set of size cuts is in Appendix D.3.

standard deviation more reserve-rich lowers its loan share by around 1.8 percent more than the average bank (column 1). The deposit-funding and bank-capitalisation interactions are small and statistically insignificant.

The quarter effects absorb common loan demand but not interbank fragmentation, since the core and periphery moved along different paths through the sovereign crisis. When these are replaced with core/periphery-by-quarter fixed effects, which absorb each region's own path, the reserve interaction strengthens rather than weakens (column 2). The interaction is therefore estimated from variation within region and quarter, and does not reflect the core/periphery divergence.

The concentration of the response among reserve-rich banks supports the opportunity-cost channel developed in the next section.

3 A macroeconomic model of credit rationing

At the core of the model is a canonical New Keynesian (NK) closed economy model. The representative household chooses consumption, C_t , labour supply, H_t , and allocates savings across bank deposits, S_t , risk-free nominal bonds, B_t , and firm equity, E_t , to maximise expected lifetime utility $\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s U(C_{t+s}, H_{t+s})$. The consumption-savings decision yields the standard Euler equation,

$$1 = \mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} \right] R_t, \quad (3.1)$$

where $\Lambda_{t,t+1} = \beta U'(C_{t+1})/U'(C_t)$ is the stochastic discount factor, R_t is the gross nominal return on deposits, and the return on bonds $R_t^B = R_t$ in equilibrium. The labour supply condition equates the real wage with the marginal rate of substitution: $W_t/P_t = -U'(H_t)/U'(C_t)$. Equity purchases are characterised by a zero-arbitrage condition outlined below.

A wholesale goods sector combines labour and purchases capital to produce a homogeneous intermediate good. This good is then sold to a retail sector under perfect competition. The retailers differentiate the intermediate good, selling their output under monopolistic competition subject to the Calvo (1983) price-setting friction.

Following Swarbrick (2023), the wholesale goods sector is characterised by information frictions in firm investment. This builds on two key assumptions. The first is that all investment projects require a single unit of productive capital for which firms need to raise a fixed quantity of external finance. This assumption ensures that firms rely on outside funding, but by treating all investment opportunities as a fixed size, we capture the inability of small firms to diversify risk.

The second key assumption is that every period, each intermediate goods firm draws a project characterised by a production technology, productivity level and a risk profile. A proportion η of firms, denoted *corporates*, have a perfectly observable project and so are suitable for raising funds via a bond market. The remaining $1 - \eta$ firms have a privately observed project, a

proportion λ of which have no idiosyncratic risk (*safe*), whereas the remaining $1 - \lambda$ have a risky project that will only succeed with probability p_t (*risky*).

Because all loans are the same size, in a decentralised bond market the interest rate would be the only screening device, and safe firms would be rationed whenever the rate exceeds their expected return. The information asymmetry gives rise to a banking sector that can outperform bond markets by screening borrowers.¹³ Despite not modelling differences in firm size, the fixed size of investment and information asymmetry capture salient features of small business: having private information and being unable to diversify risk. For this reason, we refer to the non-corporates as small firms.

Before turning to the problem facing the borrowers and lenders in the economy, we first outline details about monetary policy. We will differ from the standard NK model by giving the central bank three instruments. The first is the standard tool: the main policy target rate, R_t^p , which the central bank charges for short-term liquidity operations. Second, the central bank can directly vary the amount of liquidity to influence the short-term market interest rate on household savings, R_t .¹⁴ This can be targeted through private sector bond purchases, or direct liquidity provision to the bank sector.¹⁵ The third instrument is the interest rates on standing deposit ($\underline{R}_t$) and lending ($\bar{R}_t$) facilities. Banks can hold excess reserve balances in the former and access additional liquidity as required via the latter.¹⁶

3.1 Banks

Bank assets consist of business loans and central bank reserves, while liabilities include central bank liquidity and household deposits. The composition of the liability side is addressed below. We begin by examining the asset side. To extend finance to firms, the banks can post the terms of a menu of loans that borrowers can choose on application.¹⁷ Loans are single-period, with terms specifying both an interest rate and an approval probability. This flexibility allows banks to offer multiple loan options at different interest rates to separate borrowers. These options are designed as incentive-compatible, or self-selecting, contracts. For instance, a risky firm might accept a higher interest rate if it increases the probability of approval compared to a lower-interest loan.

Specifically, the lenders post contracts $c_t^i = \{\tau_t^i, x_t^i\}$ for $i \in \{s, r\}$, where τ_t^i is the nominal repayment rate, and x_t^i the financing, or approval probability.¹⁸ Letting R_t^i denote the gross nominal rate of return on capital for a type- i project and p_t the success probability of risky

¹³Indeed, a major rationale for the existence of banks is to mitigate borrowers' financing frictions (see Flanagan, 2025).

¹⁴The first two instruments are implicitly the same, however to capture the policy tools used in practice, we model the implementation of the first via a Taylor rule, and the second to allow shifts of the market rate below the main policy rate towards the policy floor, $\underline{R}_t$.

¹⁵Either are equivalent as in equilibrium corporate bonds, household savings, and central bank loans to the banks will clear at the same interest rate.

¹⁶There is no regulatory reserve balance and so all bank reserve balances at the central bank are excess reserves.

¹⁷Because firm-type is drawn every period, it is not possible that banks learn the firm type over time and so credit terms are not a function of a firm's history.

¹⁸The baseline contract excludes collateral. With sufficient pledgeable wealth, collateral can serve as a screening device (Bester, 1985). Appendix E extends the bank's problem to allow limited collateral, showing how it changes the threshold for credit rationing without eliminating the mechanism.

projects, lenders set contract terms subject to firms' individual rationality (IR) and incentive compatibility (IC) constraints. The relevant constraints can be written:

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} \right] \tau_t^s \leq \mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} R_{t+1}^s \right] \quad (3.2)$$

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} p_{t+1} \right] \tau_t^r \leq \mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} p_{t+1} R_{t+1}^r \right] - \mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} p_{t+1} (R_{t+1}^r - \tau_t^r) \right] \frac{x_t^s}{x_t^r}. \quad (3.3)$$

The first constraint ensures the participation of safe firms by promising weakly positive surplus. The second provides an incentive for the risky firm to reveal their type, even choosing a higher interest rate loan. This is achieved through a higher probability of loan approval. When the lending contracts separate borrowers, therefore, safe loans have low repayment rates but a lower approval rate, while risky loans have a higher repayment rate but higher approval rate.¹⁹

Free entry of banks implies a zero-arbitrage condition. Banks enter, access liquidity at rate R_t via household deposits or central bank loans, and post loan contracts. Matching to firms is proportional to bank balance sheet size. This matching assumption pins down equilibrium intermediation through the free-entry condition and generates equilibrium excess reserves, making the interest on reserves the relevant opportunity cost of lending in this model.²⁰ Other motives for holding reserves, such as liquidity risk, can also make reserve remuneration relevant to lending incentives, although the opportunity cost may then depend on additional factors.

Denoting the volume of liquidity provision for bank j as S_t^j , the total volume of loans L_t^j , and the matched firms $f_t^j \equiv f_t S_t^j / \int S_t^j dj$, the total *ex post* gross nominal rate of return to the representative bank is

$$\underline{R}_{t-1} \left(S_{t-1}^j - L_{t-1}^j \right) + [\lambda x_{t-1}^s \tau_{t-1}^s + (1 - \lambda) x_{t-1}^r p_t \tau_{t-1}^r] (1 - \eta) f_{t-1}^j Q_{t-1}^K, \quad (3.4)$$

where $\underline{R}_{t-1}$ is the interest on reserves and $(S_{t-1}^j - L_{t-1}^j)$ the excess reserves of bank j .²¹

Each borrower will use the loan to purchase a single unit of capital at price Q_t^K . Therefore, total loans extended $L_t^j \equiv (1 - \eta) (\lambda x_t^s + (1 - \lambda) x_t^r) f_t^j Q_t^K$. Denoting the loan supply-demand ratio $\phi_t^j \equiv \frac{S_t^j}{(1 - \eta) f_t^j Q_t^K}$, the zero-profit condition will determine the equilibrium volume of banking sector liquidity:

$$1 = \mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} \left(\underline{R}_t + [\lambda x_t^s (\tau_t^s - \underline{R}_t) + (1 - \lambda) x_t^r (p_{t+1} \tau_t^r - \underline{R}_t)] \frac{1}{\phi_t} \right) \right] \quad (3.5)$$

where we drop the bank index j since it follows that ϕ_t must be equal for all banks. Condition (3.5) makes clear that excess reserves are an equilibrium outcome, and follows directly from the firm-matching assumption. Since a bank's lending is limited by its firm matches, any surplus liquidity is held at $\underline{R}_t$, and free entry equates the marginal expected return, across both loans

¹⁹The implications of these contracts are discussed further in the next section.

²⁰A bank with double the liquidity matches with double the firms, so banks share the same balance-sheet composition in equilibrium and we can abstract from size. This assumption matters: if matches were independent of size, banks would enter until issued loans per bank converged to zero and would never hold surplus funds.

²¹If negative, the bank must use the lending facility at rate $\bar{R}_t$. Without liquidity risk, this never occurs in equilibrium (see Appendix C.1).

and reserves, to the deposit rate. Provided the marginal unit of liquidity is profitable, banks demand more even though it means holding low-yielding reserves; the deposit rate can therefore exceed the interest on reserves even when the sector holds excess reserves. Given that bank liabilities are risk-free deposits but assets are risky loans, it is possible for there to be *ex post* profits or losses in equilibrium. When there are profits, the household will receive a dividend, recapitalising the banks when there are losses.

Given their current level of liquidity, the banks choose the loan contract terms to maximise the expected next-period real surplus by solving:

$$\begin{aligned} \max_{c_t^s, c_t^r} \mathbb{E}_t \left[\Lambda_{t,t+1} \left\{ \lambda x_t^s \left(\frac{\tau_t^s}{P_{t+1}} - \frac{R_t}{P_{t+1}} \right) + (1 - \lambda) x_t^r \left(p_{t+1} \frac{\tau_t^r}{P_{t+1}} - \frac{R_t}{P_{t+1}} \right) \right\} \right] \\ \text{s.t. } 0 \leq x_t^s \leq x_t^r \leq 1 \\ \lambda x_t^s + (1 - \lambda) x_t^r \leq \phi_t \end{aligned} \quad (3.6)$$

and subject to constraints (3.2) and (3.3). The $x_t^r \geq x_t^s$ also follows from the IC constraints and the inequality constraint (3.6) is a loan-supply constraint. ϕ_t is determined by (3.5) and is less than one when loan demand exceeds available liquidity. Although this is not a hard constraint since banks can always access additional funds, further liquidity comes at the policy ceiling rate $\bar{R}_t$, making it unprofitable. This implies the possibility of loan-supply shortages depending on macro-financial conditions and the monetary policy stance.

When constraint (3.6) is slack, then there is excess liquidity in the banking sector, banks hold excess reserves which are held in the central bank deposit facility. The solution to the bank's problem gives

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (p_{t+1} R_{t+1}^r - \underline{R}_t) \right] = \varrho_t - \psi_t \frac{1}{1 - \lambda} + \varphi_t^r \frac{1}{1 - \lambda} \quad (3.7)$$

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} ((\lambda + (1 - \lambda) p_{t+1}) R_{t+1}^s - \underline{R}_t) \right] = \varrho_t + \varphi_t^r - \varphi_t^s, \quad (3.8)$$

where ϱ_t is the Lagrange multiplier on the feasibility constraint, φ_t^s and φ_t^r those on x_t^s and $1 - x_t^r$ respectively, and ψ_t is the Lagrange multiplier on $x_t^r - x_t^s$. These first-order conditions are also subject to complementary slackness conditions that include zero-lower bounds on the four Lagrange multipliers:

$$\varphi_t^s, \varphi_t^r, \varrho_t, \psi_t \geq 0. \quad (3.9)$$

3.2 Intermediate goods firms

Intermediate goods firms draw their project type at the end of period t and seek external finance. If they draw an observable project, then they are characterised as a *corporate* and access funds in the bond market. Otherwise, as a *small firm*, they must apply for a bank loan. They may or may not be successful in securing funds; if firms are successful, they purchase the required unit of capital at nominal price Q_t^K , ready for production in the following period, otherwise, they must exit. New firms can enter but must pay fixed entry costs.

Of the funded risky projects, a proportion $1 - p_{t+1}$ will fail in the next period before production begins, with all capital lost. Success probability, $p_{t+1} \in [0, 1]$, is subject to aggregate risk. After production, the firm repays the loan, sells the depreciated capital back to the capital market at Q_{t+1}^K , and returns any surplus to the household.

We denote corporates, safe- and risky-project firms with superscript $i \in \{c, s, r\}$. A successfully funded project requires a single unit of capital that is converted into ω_t^i productive units where $\omega_t^r = 1/p_t > \omega_t^c = \omega_t^s = 1$.²² The firm hires $h_t(\omega_t^i)$ units of labour and produces output using constant returns to scale technology,

$$y_t(\omega_t^i) = z_t [\omega_t^i]^\alpha [h_t(\omega_t^i)]^{1-\alpha}, \quad (3.10)$$

where aggregate total-factor productivity, z_t , follows a stationary stochastic process. Capital depreciates at rate δ , so the remaining capital after production will be $\omega_t^i(1 - \delta)$. Denoting the final consumption good price P_t , the nominal market value of capital Q_t^K , and the wholesale good price P_t^W , we can write the real value of a successfully funded type- i firm as:

$$V_t^i = \max_{h_t(\omega_t^i)} \left\{ \frac{P_t^W}{P_t} y_t(\omega_t^i) - \frac{W_t}{P_t} h_t(\omega_t^i) - \left(\tau_t^i \frac{Q_{t-1}^K}{P_t} - (1 - \delta) \omega_t^i \frac{Q_t^K}{P_t} \right) + \bar{V}_t \right\}, \quad (3.11)$$

where τ_t^i is the aggregate-state indexed loan repayment rate and $\bar{V}_t$ represents the *ex-ante*, real continuation value of a firm, prior to drawing its type, given by:

$$\bar{V}_t \equiv \mathbb{E}_t [\Lambda_{t,t+1} (\eta V_{t+1}^c + (1 - \eta) (\lambda x_t^s V_{t+1}^s + (1 - \lambda) x_t^r p_{t+1} V_{t+1}^r))], \quad (3.12)$$

which is a probability-weighted average of future values conditional on the next period project draw. The solution to (3.11) yields the standard labour demand condition equating the real wage with the marginal product of labour. Since output per worker and the efficiency capital-labour ratio are equal across firm types, we can write the gross nominal return on capital as

$$R_t^i \equiv \frac{\alpha \frac{P_t^W}{P_t} y_t^i + (1 - \delta) q_t^K \omega_t^i}{q_{t-1}^K} \Pi_{t-1,t}, \quad (3.13)$$

where $q_t^K \equiv Q_t^K / P_t$ is the real price of capital. Noting that the gross return on efficiency units of capital, $\alpha \frac{y_t(\omega_t^i)}{\omega_t^i} + (1 - \delta)$, is equal for all firms, it follows that $R_t^r = \omega_t^r R_t^c = \omega_t^r R_t^s$.

3.2.1 Firm entry

Firms can earn positive profits due to the information asymmetry and because each firm is a fixed size, the number of firms in the economy will matter for aggregate outcomes. To model firm dynamics, we assume firms pay a small fixed cost equal to F units of the final good to enter. The entry cost is a fee paid to households (a rebate), so it does not absorb real resources and enters household income as a lump-sum transfer.

To pay the entry costs, firms sell equity to households at nominal price Q_t^E . Total equity

²²The assumption that $\omega_t^r = 1/p_t$ is not required but allows us to isolate the effects of risk.

purchases, E_t , corresponds to the number of new firms invested in, where each share is a claim on the future profit streams of an entering firm.

$$E_t = (f_t - (\eta + (1 - \eta) (\lambda x_{t-1}^s + (1 - \lambda) x_{t-1}^r)) f_{t-1}).$$

Under this assumption, new firms will enter until the expected discounted profits $\bar{V}_t$, given by equation (3.12), equals the entry cost F .

3.3 Retailers and capital good producers

Firms in the retail sector differentiate a homogeneous intermediate good purchased at price P_t^W into final retail goods which are sold under monopolistic competition subject to a Calvo friction, whereby they face a fixed probability, ξ of being able to update each period. This leads to the usual conditions for the optimal price for price setters, P_t^* , and law of motion for inflation $\Pi_{t-1,t} \equiv P_t/P_{t-1}$, which are characterised by the usual NK Phillips curve.²³

We also introduce capital producers that invest in new capital and buy and sell in a market for capital goods at the end of the period. Firms that secure funding seek to purchase enough capital to make up the required amount and exiting firms sell their remaining capital stock. Subject to the demand for capital and the market price, Q_t^K , capital producers purchase final retail goods, converting them into new capital goods to sell to firms under perfect competition. We assume that it is costly to adjust investment plans following Christiano et al. (2005). Because $\omega_t^r = 1/p_t$ and $\omega_t^c = \omega_t^s = 1$, in aggregate, the total units of capital sold is equal to the efficiency units in production, and therefore the aggregate capital stock evolves according to:

$$K_t = (1 - \delta)K_{t-1} + I_t \left[1 - \frac{\phi_K}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right]. \quad (3.14)$$

3.4 Market clearing and aggregation

Labour market clearing implies that total labour demanded by the three types of firm will equal the labour supplied by households, H_t . The perfect labour market implies that all firms will choose the same efficiency-capital-labour ratio and so, giving the capital markets clearing condition as

$$K_t \equiv [\eta + (1 - \eta) (\lambda x_t^s + (1 - \lambda) x_t^r)] f_t, \quad (3.15)$$

we can write the aggregate labour demand and intermediate good production function as

$$\begin{aligned} \frac{W_t}{P_t} &= (1 - \alpha) MC_t z_t \left(\frac{K_{t-1}}{H_t} \right)^\alpha \\ Y_t^W &\equiv \int_i y_t^i di = z_t K_{t-1}^\alpha H_t^{1-\alpha} \end{aligned}$$

²³See Appendix A for equilibrium conditions.

where $MC_t \equiv P_t^W/P_t$ is the real marginal cost in the retail sector. Finally, the goods market clears, implying the aggregate resource constraint $Y_t = C_t + I_t$ where $Y_t = Y_t^W/\Delta_t$, and Δ_t is a measure of price dispersion.

3.5 Monetary policy

Unlike the standard New Keynesian economy, the volume of liquidity in circulation affects the market interest rate through bank balance sheet and profitability constraints. When the central bank purchases corporate bonds, it replaces higher-return assets held by the private sector with lower-return reserves held by banks; the bond market then only clears at a lower interest rate.²⁴ The central bank can therefore target the short-term market rate, R_t , through open market operations.

To distinguish between the central bank interest rates, we assume that the main policy interest rate is set to satisfy a Taylor-type rule:

$$\log(R_t^p) = \log\left(\frac{\Pi}{\beta}\right) + \gamma_\pi(\pi_{t-1,t} - \pi^*) + \gamma_y(y_t - y^*) \quad (3.16)$$

where $\pi_{t-1,t} \equiv \log(\Pi_{t-1,t})$ and $y_t \equiv \log(Y_t)$. In addition to the main policy rate, the central bank has two further tools. The first is the interest rate on excess reserves, set equal to the main policy rate minus a desired corridor: $\underline{R}_t = R_t^p - c_t$. The second is liquidity operations that move the market rate below the main policy rate towards the floor. We characterise this departure as a quantitative easing policy: $R_t = R_t^p - q_t$. The spread between the market rate and the floor is then $R_t - \underline{R}_t = c_t - q_t$, equal to the corridor when liquidity is not eased ($q_t = 0$) and compressing below it under quantitative easing.

4 Monetary Policy Transmission: Mechanisms and Quantitative Results

This section characterises how bank lending incentives depend jointly on macro-financial conditions and monetary policy. These incentives give rise to three equilibrium lending regimes, distinguished by whether all loan applications are financed, equilibrium loan supply falls short of demand, or banks actively ration credit.

A key implication is that liquidity policies operate through two competing channels: a standard expansionary demand channel and a contractionary credit-rationing channel that works through banks' lending incentives. These map to the funding-cost and opportunity-cost channels of Section 2, now in general equilibrium. Which channel dominates depends on risk and profitability conditions, and policy can shift the economy across regimes.

We first characterise equilibrium loan contracts and identify the three regimes. We then illustrate the steady-state thresholds using comparative statics, and finally quantify the competing channels using impulse responses simulations.

²⁴Equivalently, the central bank can provide liquidity directly via refinancing operations. In equilibrium, the interest on central bank loans equals the rate on corporate bonds and household savings.

4.1 Loan contracts and equilibrium regimes

The bank first-order conditions (3.7) and (3.8) together with the (binding) IC and IR constraints, (3.3) and (3.2) imply a menu of loan contract specifying the loan repayment, τ_t^i , and loan approval probability, x_t^i .

Proposition 1 *If $x_t^s > 0$ ($\varphi_t^s = 0$) then $x_t^r = 1$ ($\varphi_t^r > 0$) in equilibrium.*²⁵

Unless there is full rationing of safe firms, the approval rating on risky loans is 1 and all risky firms are funded. It follows that when banks wish to tighten lending standards due to changing macro-financial conditions, increased default risk for instance, they do so on the price of risky loans and approval probability on safe loans. Credit rationing therefore occurs on safe loans (good types) rather than risky loans (bad types), as the cost of separating the types rather than the untargeted rationing of Stiglitz and Weiss (1981).²⁶

This prediction finds support in the data: conditional on observables, rejection rates for UK SMEs rose significantly for low and average risk firms during periods of credit tightening, whereas firms rated above average risk did not face significantly higher rejection.²⁷ The screening margin should be distinguished from the response to observable risk. Where banks can rate applicants on observables, they reject those observed to be riskier, as under the lender cutoff rule in Berg (2018). Both margins operate in practice, with banks rejecting observably poor applicants, while the approval probability separates those they cannot tell apart. Because all project types deliver the same expected productive units ($\omega_t^r = 1/p_t$), aggregate outcomes depend on the total mass of firms financed rather than on which type is rationed.

The bank first-order conditions, together with the firm and bank entry conditions, determine equilibrium liquidity and lending. Three regimes can arise in this environment: all firms obtain finance; lending is restricted by a binding loan-supply constraint; or banks ration credit because the marginal information rent to risky firms offsets the marginal revenue from safe loans. The following proposition characterises these regimes.

Proposition 2 *There is no credit rationing in equilibrium if*

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (\lambda (R_{t+1}^s - \underline{R}_t) - (1 - \lambda) (1 - p_{t+1}) R_{t+1}^s) \right] > 0 \quad (4.1)$$

Using this threshold, we can identify three distinct regimes:

1. **Pooling:** *there is full approval under a pooling equilibrium ($\tau_t^s = \tau_t^r$ and $x_t^s = x_t^r = 1$) in which all firm financing needs are met if 4.1 is satisfied and:*

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} [1 - (1 - \lambda) (1 - p_{t+1})] R_{t+1}^s \right] \geq 1 \quad (4.2)$$

²⁵All proofs in Appendix C.

²⁶In Stiglitz and Weiss (1981) the bank cannot sort applicants and denies a subset at the bank-optimal rate, with the safe borrowers priced out rather than separated. Here the menu separates the types, the approval probability taking the screening role that collateral takes in Bester (1985), which our firms lack the wealth to post (footnote 18).

²⁷Armstrong et al. (2013) find this for term loans over 2001–12 using the UK Survey of SME Finances, and Swarbrick (2023) finds the same pattern in SME Finance Monitor data for 2011–17.

2. **Low-risk separating:** there is restricted lending due to a binding loan-supply constraint under a separating equilibrium ($\tau_t^s < \tau_t^r$ and $x_t^s < x_t^r = 1$) in which not all firms are financed if 4.1 is satisfied and:

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} [1 - (1 - \lambda)(1 - p_{t+1})] R_{t+1}^s \right] < 1 \quad (4.3)$$

3. **High-risk separating (credit rationing):** there is restricted lending due to credit rationing if

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (\lambda (R_{t+1}^s - \underline{R}_t) - (1 - \lambda)(1 - p_{t+1}) R_{t+1}^s) \right] = 0 \quad (4.4)$$

Corollary 1 *The incentive to ration credit rises in: the share of risky firms, $1 - \lambda$; the interest rate on reserves, $\underline{R}_t$; and the expected risky firm default rate, $1 - p_t$. Given policy rates, the credit rationing incentive always falls in the return on capital, R_{t+1}^s , if $\lambda > (1 - \lambda)(1 - p_t)$.²⁸*

Corollary 2 *Under non-rationing, the likelihood of constrained loan supply rises in the expected risky firm default rate, $1 - p_t$, and the firm share of risky firms, $1 - \lambda$*

To interpret the rationing condition (4.4), first note that from IC constraint (3.3), the information rent earned by risky firms is

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (1 - p_{t+1}) R_{t+1}^s \right] x_t^s, \quad (4.5)$$

which increases in the number of safe loans made. Condition (4.4) equates the marginal revenue from issuing a safe loan, $\lambda (R_{t+1}^s - \underline{R}_t)$, with the marginal cost from the resulting increase in the information rent. Once this cost outweighs the revenue, due to high default risk or a high return on reserves, banks ration credit until the two are equal. The interest on reserves therefore determines the opportunity cost of lending: as it rises, the incentive to ration credit increases and bank holdings of excess reserves rise.

Note that credit rationing is a profit-maximising response, not a consequence of insufficient funding. Banks can always expand their balance sheets by accessing additional liquidity; lending is tightened because the return on reserves rises relative to the return on lending. When lending margins are compressed, banks optimally reallocate towards reserves, so aggregate bank liquidity may increase even as the volume of credit extended to firms declines.²⁹

4.2 Comparative statics

Figure 2 provides a steady-state illustration of the lending regimes characterised in Proposition 2. It shows how equilibrium outcomes vary with borrower risk and the width of the policy

²⁸This holds if the number of safe firms in the economy outnumber the number defaulting risky firms. This condition will be satisfied under any reasonable calibration.

²⁹A reduction in lending does not translate one-for-one into an increase in reserves: substituting towards lower-yielding reserves reduces bank profitability and limits balance-sheet expansion. Excess reserves can also arise in low-risk environments when low firm profitability discourages entry and reduces loan demand. In that case, excess liquidity reflects weak demand rather than impaired loan supply.

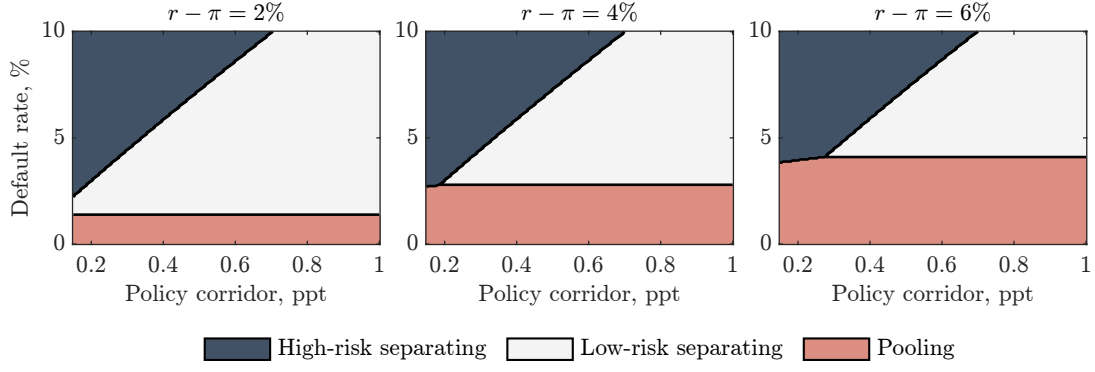


Figure 2: Comparative statics showing the role of risk (risky firm default rate), the policy corridor and the steady-state real interest rate. Other parameter values: $\eta = 0.5$, $\lambda = 0.775$, $F = 0.15$, $\beta = 0.9951, 0.9902, 0.9855$.

corridor, defined as the spread between the main policy rate and the interest on reserves. Movements along either dimension shift banks' lending incentives by altering the profitability of extending loans relative to holding reserves, generating transitions between the pooling, low-risk separating, and high-risk separating regimes.

In addition to borrower risk and the policy corridor, the three regions depend only on a small set of structural parameters. The composition of small firms between safe and risky types, $1 - \lambda$, is central: a higher share of risky firms worsens adverse selection and increases the likelihood of credit rationing. The remaining parameters are the discount factor, β , and firm entry costs, F , both of which govern the demand for credit. As illustrated across the panels of figure 2, higher β (or lower F) raises the present value of future firm profits, encouraging entry and increasing loan demand. This makes separating regimes more likely, shrinking the region in which a pooling equilibrium emerges.³⁰

Figure 3 illustrates how the regimes map into macro-financial outcomes. Although approval rates are lower in the separating regimes than under pooling, macroeconomic performance need not be worse: in the pooling region, weak firm profitability and subdued entry can generate lower steady-state output and a larger credit wedge, defined as the spread between the return on capital and the risk-free rate, despite full loan approval. The credit wedge is minimised at the boundary where condition (4.1) is satisfied and the pooling condition (4.2) binds.

Monetary policy plays an asymmetric role across these margins. By lowering the policy floor, policymakers can directly shift the credit rationing threshold outward, as shown in figure 3. However, the steady-state supply and demand for loans are determined by secular factors, including the natural rate of interest and firm entry costs, and shortfalls in loan supply cannot be eliminated through interest rate policy alone.

³⁰Secular declines in real interest rates therefore influence steady-state lending conditions, not only through standard intertemporal substitution, but also by affecting firm entry. When risk is high and lending conditions already tight, lower real rates can tighten lending further.

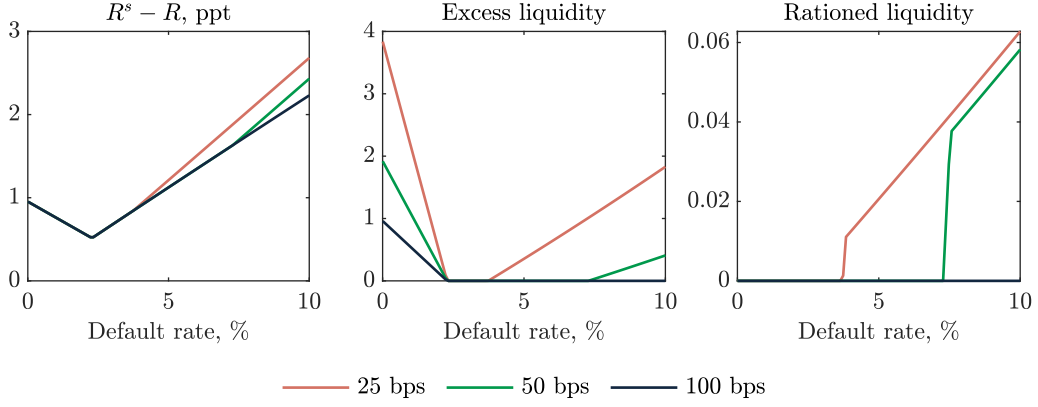


Figure 3: Comparative statics showing the role of risk (risky firm default rate) and the policy corridor on key macro-financial outcomes. Parameter values: $\eta = 0.5$, $\lambda = 0.775$, $F = 0.15$, $\beta = 0.992$.

4.3 Liquidity policies: two competing channels

We now study the transmission of liquidity policies in the presence of the three lending regimes. We model quantitative easing (QE) as a liquidity injection that compresses the spread to the floor. Because the market rate is pinned by the volume of liquidity through (3.5), parametrising the programme by its effect on the spread is equivalent to specifying the volume of liquidity directly.³¹ In the baseline simulations, the policy is calibrated to give an unexpected, temporary reduction in R_t of 25 basis points (annualised), corresponding to a move halfway from the policy rate (R_t^p) to the floor ($\underline{R}_t$), and following an AR(1) process with persistence of 0.75. Although this resembles a conventional rate cut from the perspective of households, because it lowers the market interest rate, it also alters banks' incentives by compressing the spread between the return on lending and the interest on reserves.

In the model, QE therefore operates through two competing channels. The first is the standard *expansionary* demand channel: a fall in R_t raises consumption and increases the incentive to invest. The second is a *contractionary* credit-rationing channel: by moving R_t closer to $\underline{R}_t$, the policy increases the opportunity cost of funding loans (relative to holding reserves), lowering banks' lending incentives and worsening the adverse-selection friction. When this channel dominates, banks ration credit, optimally reducing approval rates for safe loans. This causes credit spreads to rise, and investment falls even though household demand strengthens.

To evaluate the dynamic implications of the policy, we simulate the model using the King, Plosser and Rebelo (1988, KPR) class of utility with external habits in consumption and a standard New Keynesian calibration.³² For the new parameters, we set $\eta = 0.5$ so half of firms are corporates, $\lambda = 0.775$, $F = 0.15$, and unless otherwise stated, a steady-state annual risky-firm default rate of 6% ($\bar{p} = 0.985$). We focus on the role of borrower risk and the size of the liquidity programme, since these determine whether the economy crosses into the credit-rationing regime.

³¹This relationship is consistent with the empirical response of short-term market rates to asset purchase programmes; see Figure 1. The model captures this short-rate and reserve-side channel of purchases, abstracting from the duration and portfolio-balance effects on long-term yields.

³²Full details of the calibration are shown in Table 3 in Appendix A.

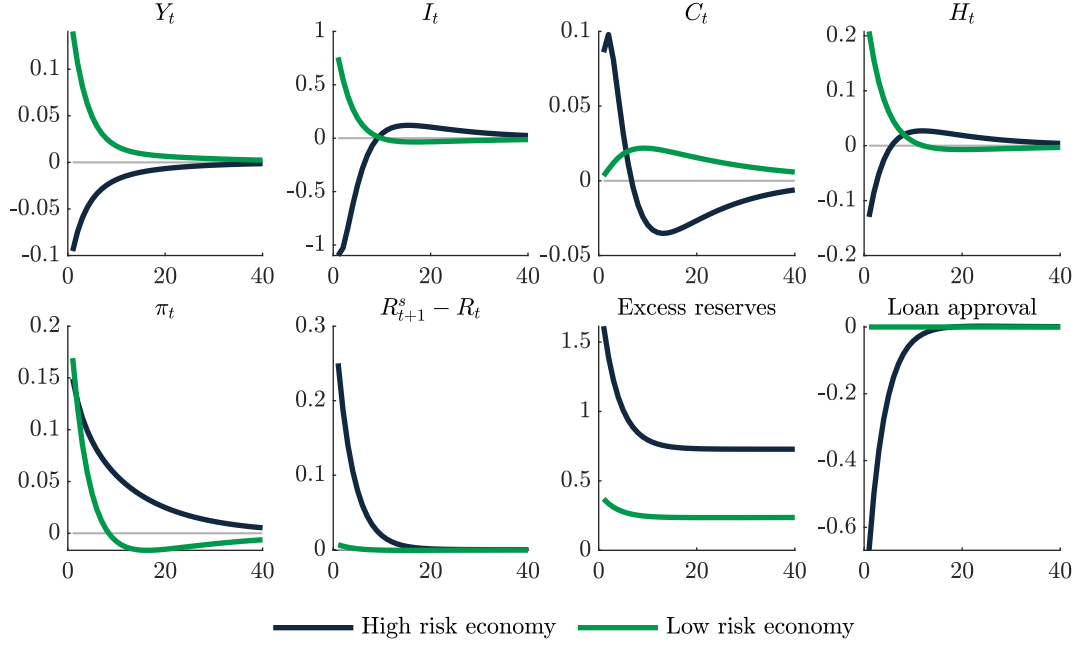


Figure 4: Impulse response functions to an unexpected temporary ‘QE’ programme. Plots in the top row show % deviation from steady state. The bottom row shows ppt deviation except for excess reserves with plots the level response. Excess reserves is given as the ratio of reserves to stock of loans. Per annum risky firm default rate is 12% (high risk) and 2% (low risk).

Figure 4 compares impulse responses to the same QE shock in two economies that differ only in steady-state borrower risk. In the low-risk economy, the programme behaves like a conventional expansionary monetary policy shock. A lower R_t raises aggregate demand through the usual intertemporal-substitution channel, increasing consumption and investment. Lending standards remain unchanged because the economy stays away from the rationing threshold, so the demand channel dominates.³³

In the high-risk economy, the same policy produces the opposite short-run output response. Because banks are already within the rationing threshold (4.1), compressing the spread between R_t and $\underline{R}_t$ strengthens the incentive to hold reserves rather than expand lending. Banks tighten lending standards by reducing safe-loan approval, credit spreads rise, and investment contracts. Reserves increase sharply, reflecting the endogenous reallocation of bank balance sheets towards low-return liquid assets. The economy therefore displays strongly state-dependent transmission: the same liquidity operation is expansionary when risk is low, but can be contractionary when risk is high because it pushes lending incentives into the rationing regime.

Figure 5 illustrates the same mechanism by varying the size of the QE programme. Small programmes leave the economy in the non-rationing region, so responses scale approximately proportionally with shock size. Larger programmes, however, move R_t sufficiently close to $\underline{R}_t$ that condition (4.1) binds, triggering a regime shift: loan approval falls, spreads rise endogenously, and output can contract on impact despite higher consumption demand. Programme size therefore matters nonlinearly.

³³ A subtle point is that while the compressed spread moves the economy closer to the credit-rationing threshold, it simultaneously moves it away from the loan-supply constraint. Reserves therefore rise endogenously as lending expands, reflecting lending profitability rather than any constraint on bank funding.

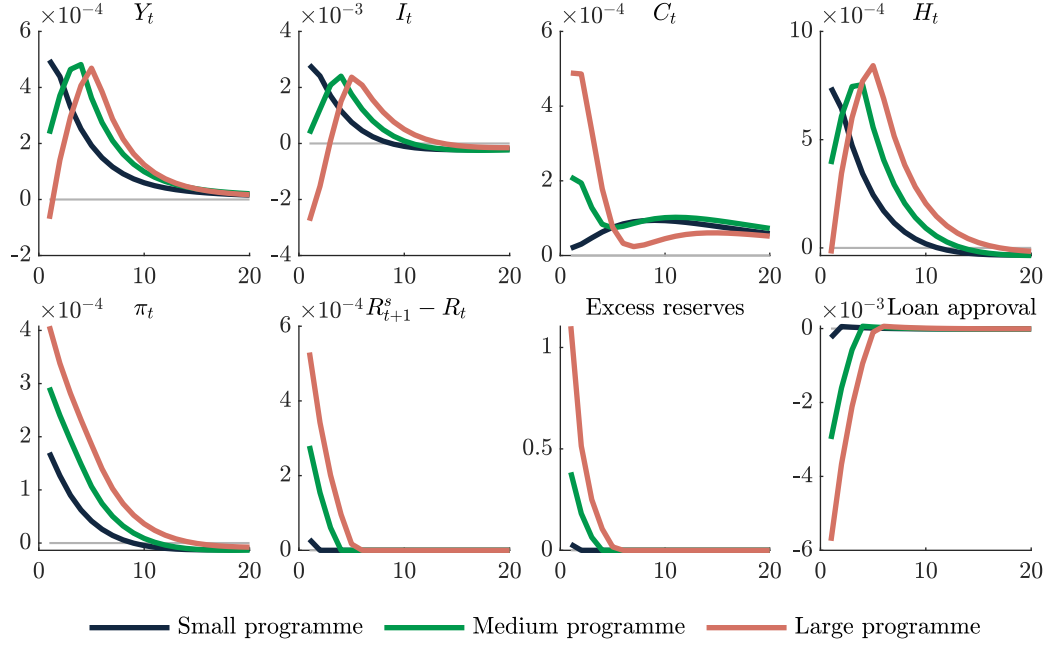


Figure 5: Impulse response functions to a temporary ‘QE’ programme. Plots in the top row show % deviation from steady state. The bottom row shows ppt deviation except for excess reserves which plots the ratio of reserves to stock of loans.

4.4 Robustness: investment adjustment costs and preferences

We examine how investment adjustment costs and alternative preferences affect the balance between the expansionary demand channel and the contractionary credit-rationing channel. Across the specifications considered, liquidity provision worsens small-business lending conditions, but its net effect on output depends on the responsiveness of investment and consumption. For example, introducing investment adjustment costs dampens the contractionary response of investment to tighter lending conditions, allowing the expansionary demand channel to dominate even in high-risk economies.³⁴

Another margin is the strength of the wealth effect on labour supply. Adopting preferences of the type proposed by Jaimovich and Rebelo (2009, JR), which allow for a weak short-run wealth effect on labour supply, can substantially dampen the consumption response to lower interest rates. As a result, the expansionary demand channel is weakened. In this case, liquidity programmes can generate much larger and more persistent contractions in output, as the decline in investment driven by credit rationing is no longer offset by higher consumption.³⁵

Finally, combining JR preferences with investment adjustment costs reverses this result. Adjustment costs limit the contractionary investment response induced by tighter lending standards, while the weak wealth effect prevents excessive labour supply responses. Together, these forces restore the dominance of the demand channel, so that even large liquidity programmes generate

³⁴This mechanism is illustrated in Figure 7 in Appendix F, which shows that with adjustment costs, the same QE programme generates a positive output response despite tighter lending standards.

³⁵See Figure 8 in Appendix F. Working in the same direction, increasing the degree of habit formation with the original KPR preferences will also dampen the offsetting effects of consumption demand. However, quantitatively, the effect is insufficiently strong to alter the key results.

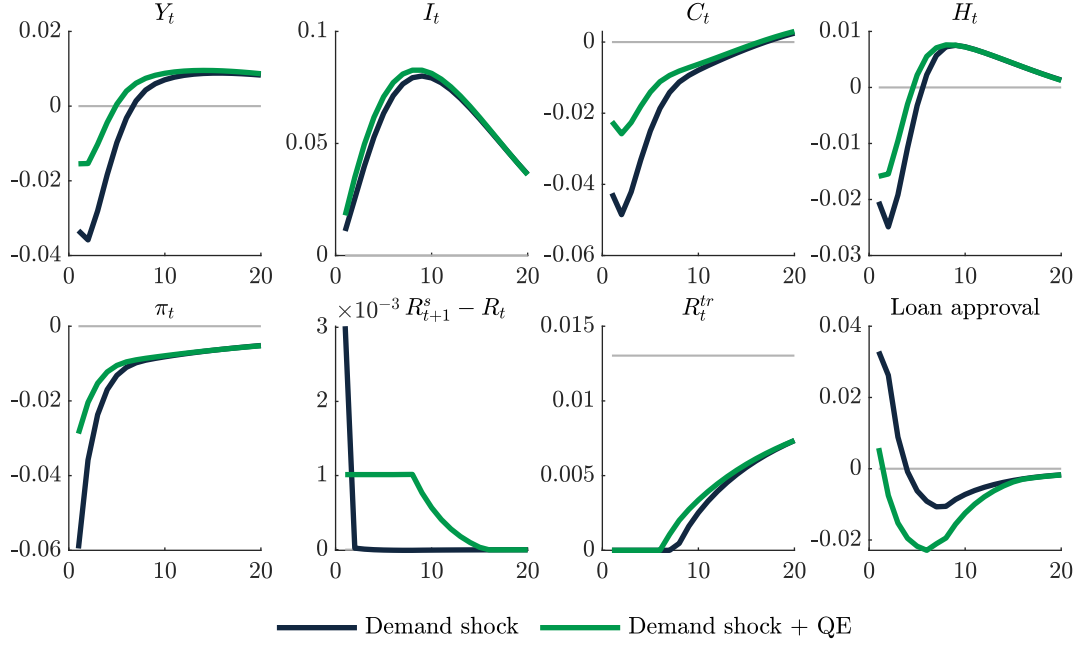


Figure 6: Impulse response functions to a demand shock with and without a ‘QE’ response and an economy with investment adjustment costs ($\phi_K = 4$). Plots in the top row show % deviation from steady state. The bottom row shows ppt deviation except for excess reserves which plots the ratio of reserves to stock of loans.

positive output responses despite increased credit rationing.³⁶

4.5 Liquidity policies at the effective lower bound

In practice, liquidity policies are most often used when the main policy interest rate, R_t^p , is constrained by the effective lower bound (ELB). This raises the question of how the ELB alters the balance between the expansionary demand channel and the contractionary credit-rationing channel. The key mechanism in this environment is the response of inflation expectations.

Figure 6 plots impulse response functions to a negative demand shock, with and without a QE response, in an economy with investment adjustment costs.³⁷ The demand shock is modelled as an unexpected increase in the household discount factor, followed by its gradual return to steady state.³⁸

The shock pushes the policy rate to the ELB, where it remains for several periods. In the absence of QE, the inability of monetary policy to offset the decline in demand leads to a sharp fall in inflation. Higher household saving increases bank liquidity and results in a temporary easing of lending standards, but this effect is insufficient to stabilise aggregate activity.

When QE is implemented alongside the demand shock, the central bank expands reserve bal-

³⁶See Figure 9 in Appendix F. The balance between these forces is parameter-dependent: estimated DSGE models typically feature substantial investment adjustment costs (e.g. Smets and Wouters, 2003, 2007), which favour the demand channel, but estimated short-run wealth effects on labour supply tend to be small (e.g. Galí et al., 2012; Schmitt-Grohé and Uribe, 2012), which favours the credit-rationing channel.

³⁷Investment adjustment costs are included to ensure that the shock generates a contraction in aggregate demand. Without adjustment costs, households would substitute away from consumption towards investment and labour, partially offsetting the decline in demand.

³⁸The shock increases β by 0.01, and follows an AR(1) process with persistence 0.9.

ances, lowering the overnight market rate below the policy rate and all the way to the policy floor.³⁹ As in the previous section, this activates two competing channels: an expansionary demand channel and a contractionary credit-rationing channel. Crucially, however, both channels are inflationary.

As shown analytically in the next section, tighter lending conditions raise firms' marginal costs, introducing a cost-push effect. Even when credit standards tighten and investment is partially restrained, inflation expectations rise. While the policy rate is constrained at the ELB, higher expected inflation lowers real interest rates, reinforcing the expansionary demand channel. QE therefore remains expansionary on net despite tighter lending standards, and policy lift-off occurs earlier.

5 Optimal monetary policy design

This section studies the implications of credit frictions for welfare and optimal monetary policy in a simplified environment. We derive the log-linear New Keynesian representation in which endogenous lending distortions enter as a credit wedge, and obtain the associated quadratic welfare loss function. Building on this structure, we study optimal monetary policy through targeting rules under discretion and commitment, and examine the role of liquidity policy when nominal interest rates are constrained by the effective lower bound. The analysis highlights how credit frictions act as endogenous cost-push shocks and alter the inflation–output trade-off faced by the central bank.

Our analysis differs from two strands of work on optimal policy with credit frictions. Where an endogenous spread acts as a cost-push distortion that policy accommodates (Cúrdia and Woodford, 2010, 2016; Carlstrom et al., 2010; De Fiore and Tristani, 2013), here we will find that easing is moderated by the incentive to ration. Where a second instrument is assigned to financial shocks and quantitative easing is generally expansionary (Sims et al., 2023; De Fiore and Tristani, 2019), here liquidity policy is regime-dependent and can be contractionary.

5.1 Restricted environment and the efficient allocation

Assumption 1 (Restricted version) *We consider a restricted version without physical capital and with a fixed number of firms, f . Firms require external finance to operate, but we abstract from entry dynamics in order to focus on the allocation and policy implications of credit distortions. Preferences are additive-separable in consumption and labour:*

$$U(C_t, H_t) = \ln(C_t) - \varphi_H \frac{H_t^{1+\varphi}}{1+\varphi} \quad (5.1)$$

Additionally, the following parameter restrictions are imposed: $\eta = \pi^ = 0$.*⁴⁰

³⁹As before, the policy is implemented as a direct reduction in the market interest rate, here falling by 50 basis points relative to the policy rate, with persistence 0.8.

⁴⁰The key mechanism, endogenous credit rationing driven by adverse selection and the opportunity cost of reserves, is preserved, and the credit wedge conditions inherit the same qualitative properties as the full model.

The social planner's problem is to choose allocations C_t , H_t and A_t to maximise household utility $\sum_t \beta^t U(C_t, H_t)$ subject to the resource constraint $C_t = Y_t$ and technology $Y_t = A_t H_t^{1-\alpha}$, where $A_t = \exp(z_t) [f x_{t-1}]^\alpha$, and where $x_t \equiv \lambda x_t^s + (1 - \lambda) x_t^r$ is the total share of projects that are funded.⁴¹

The solution to the social planner problem yields $x_t = 1$ and $U_{H,t}/U_{C,t} = -(1 - \alpha)A_t H_t^{-\alpha}$. This is equal to a full-lending (first-best), flexible-price economy where labour subsidies are imposed to remove the steady-state distortion stemming from monopolistic competition. In the optimal policy exercises that follow, we include constant labour subsidies and lending subsidies to impose the efficient steady-state allocation.⁴²

5.2 The New Keynesian model with a credit wedge

Taking a first-order approximation around the efficient allocation yields the New Keynesian IS relationship and Phillips curve:

$$\pi_t = \beta \mathbb{E}_t [\pi_{t+1}] + \kappa \tilde{y}_t - \kappa \alpha \ln(x_{t-1}) \quad (5.2)$$

$$\tilde{y}_t = \mathbb{E}_t [\tilde{y}_{t+1}] - (r_t - \mathbb{E}_t [\pi_{t+1}] - r_t^n) \quad (5.3)$$

where

$$r_t^n = \rho + \mathbb{E}_t [\Delta z_{t+1}], \quad \kappa = \frac{(1 - \xi)(1 - \xi\beta)}{\xi} \frac{1 + \varphi}{1 - \alpha} \quad (5.4)$$

and $\tilde{y}_t$ is the efficient output gap, π_t the log of inflation, and r_t the log nominal market interest rate.⁴³

The key innovation relative to the benchmark New Keynesian model (e.g. Clarida et al., 1999) is that the marginal cost depends endogenously on credit allocation. When lending is constrained, effective productivity falls, generating a cost-push term in the Phillips curve. This mechanism mirrors the regime structure analysed earlier. Loan-supply constraints and credit rationing reduce the funded share of projects, generating a supply-side distortion that feeds directly into inflation dynamics.⁴⁴

5.2.1 Two policy instruments and two channels

In the full model the policymaker controls the main policy rate r_t^p , the market rate r_t , and the interest on reserves $\underline{r}_t$. Since the main policy rate has no independent allocative role, in this section the policymaker sets r_t directly, leaving two instruments, r_t and $\underline{r}_t$, with the spread to the floor $\nu_t \equiv r_t - \underline{r}_t$ coinciding with the corridor. These two instruments affect lending through

⁴¹Since there are many projects and each has the same expected return, a pool of risky projects has the same net present value as a pool of safe projects. Therefore, we do not discriminate and can focus only on x_t .

⁴²Equilibrium conditions and details of the steady-state lending subsidy are provided in Appendix B.

⁴³Further details in Appendix B.

⁴⁴Since $\ln(x_{t-1}) \leq 0$, the credit wedge acts as a one-sided cost-push distortion: for a given efficient output gap, constrained lending raises inflation relative to the full-lending case.

distinct channels. The log-linearised bank first-order condition under rationing yields:⁴⁵

$$\ln(x_t) = \Lambda_0 - \Lambda^D (r_t - \rho) + \Lambda^\nu \hat{\nu}_t + \Lambda_y \mathbb{E}_t [\tilde{y}_{t+1}] + \Lambda_\pi \mathbb{E}_t [\pi_{t+1}] + u_{r,t} \quad (5.5)$$

where $\hat{\nu}_t \equiv \nu_t - \nu$ denotes the deviation of the spread from steady state, $u_{r,t}$ collects exogenous shocks, and:

$$\Lambda^D \equiv \frac{1 - \alpha}{1 + \alpha\varphi} \frac{R^s}{R^s - 1}, \quad \Lambda^\nu \equiv \frac{1 - \alpha}{1 + \alpha\varphi} \frac{\lambda R}{(R^s - 1)[\lambda - (1 - p)(1 - \lambda)]}. \quad (5.6)$$

The spread ν_t enters the linearised system as a convenient summary of the floor rate's effect on lending incentives, consistent with the empirical evidence in Section 2, where conditioning on the market rate isolates the floor as the operative policy variable. The two coefficients capture the two competing channels of monetary policy on lending identified in Section 4.3. The demand channel, governed by $\Lambda^D > 0$, captures how a lower interest rate raises aggregate demand and the expected return on lending, increasing banks' willingness to lend. The credit-rationing channel, governed by $\Lambda^\nu > 0$, captures how a narrower spread to the floor increases the opportunity cost of lending, reducing banks' lending incentive.

Lending is determined by three inequality constraints corresponding to the regimes identified in Proposition 2:⁴⁶

$$\ln(x_t) \leq \Lambda_0 - \Lambda^D (r_t - \rho) + \Lambda^\nu \hat{\nu}_t + \Lambda_y \mathbb{E}_t [\tilde{y}_{t+1}] + \Lambda_\pi \mathbb{E}_t [\pi_{t+1}] + u_{r,t} \quad (\text{Rationing}) \quad (5.7)$$

$$\ln(x_t) \leq \Upsilon_0 - \Upsilon_r (r_t - \rho) + \Upsilon_y \mathbb{E}_t [\tilde{y}_{t+1}] + \Upsilon_\pi \mathbb{E}_t [\pi_{t+1}] + u_{l,t} \quad (\text{Loan supply}) \quad (5.8)$$

$$\ln(x_t) \leq 0 \quad (\text{Feasibility}) \quad (5.9)$$

Banks ration credit in the high-risk separating regime when (5.7) binds; loan supply falls short of demand in the low-risk separating regime when (5.8) binds; and there is full lending under pooling when only the feasibility constraint (5.9) binds.⁴⁷

Proposition 3 (No-ELB benchmark) *Away from the effective lower bound on the floor, the policymaker can always set ν_t sufficiently large to ensure that credit rationing does not occur. The spread instrument ν_t neutralises the credit-rationing channel, though lending may still be constrained by low bank profitability through the liquidity channel.*⁴⁸

Proposition 3 establishes that the spread to the floor is sufficient to address credit rationing but not constrained lending more broadly. When $r_t \geq \underline{r}^{ELB}$ binds, $\nu_t = r_t - \underline{r}^{ELB}$ and the spread is no longer a free instrument. The two channels become yoked: lowering r_t both stimulates demand and compresses ν_t , which can worsen banks' lending incentive. The credit wedge then depends on r_t through the net coefficient $\Lambda_r \equiv \Lambda^D - \Lambda^\nu$. Its sign depends on the point around

⁴⁵In general, the condition for Λ^ν should also include the lending subsidy, however, in all reasonable parameterisations, a subsidy is not required for $x = 1$ and so we drop this here for brevity.

⁴⁶See Appendix B for further details.

⁴⁷Since there are no excess reserves in the low-risk separating regime, the interest on reserves is irrelevant to bank profitability, which is why $\hat{\nu}_t$ does not appear in (5.8). The spread instrument $\hat{\nu}_t$ therefore affects only the rationing margin, consistent with Corollary 1.

⁴⁸All proofs in Appendix C.

which the lending condition is approximated. At the efficient steady state, $\Lambda_r > 0$, reflecting the local dominance of the demand channel. At the boundary of the rationing regime, $\Lambda_r = 0$, while an approximation within the credit-rationing regime yields $\Lambda_r < 0$, reflecting the dominance of the credit-rationing channel.⁴⁹

5.3 Welfare loss function and optimal targeting rules

Taking a second-order approximation to household utility around the efficient undistorted steady state yields the loss function:⁵⁰

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(-\lambda_x \frac{1-\alpha}{1+\varphi} \alpha \ln(x_{t-1}) + \frac{1}{2} \left[\pi_t^2 + \lambda_x (\tilde{y}_t - \alpha \ln(x_{t-1}))^2 \right] \right) \quad (5.10)$$

where

$$\lambda_x = \frac{1+\varphi}{1-\alpha} \frac{(1-\beta\xi)(1-\xi)}{\zeta\xi} = \frac{\kappa}{\zeta} \quad (5.11)$$

Without the credit friction, $\ln(x_{t-1}) = 0$ and the loss function collapses to the standard case, which is quadratic in inflation and the output gap. The possibility of credit rationing introduces additional sources of welfare loss.

First, reduced lending lowers effective productivity and output. This appears as a linear welfare loss through the term in $\ln(x_{t-1})$, capturing the direct cost of financing distortions. Second, there is a quadratic term in $\tilde{y}_t - \alpha \ln(x_{t-1})$, which is the gap between output and its natural (flex-price) level taking the credit allocation as given. To see this, note that real marginal cost can be written as

$$\widehat{mc}_t = \frac{1+\varphi}{1-\alpha} (\tilde{y}_t - \alpha \ln(x_{t-1})) \quad (5.12)$$

so that $\tilde{y}_t - \alpha \ln(x_{t-1}) = 0$ when prices are flexible. This gap enters the quadratic welfare term, while the separate linear term captures the cost of constrained lending itself. Under credit rationing, stabilising marginal cost requires output below its efficient level: closing the efficient output gap ($\tilde{y}_t = 0$) would instead leave marginal costs elevated, generating inflation. The loss function thus captures a trade-off between stabilising inflation and closing the efficient output gap that is absent under full lending.

5.4 Optimal policy under discretion

Under discretion, the policymaker minimises period welfare loss subject to the Phillips curve (5.2), the credit wedge conditions (5.7)–(5.9), and the effective lower bound on the floor, $\underline{r}_t \geq \underline{r}^{ELB}$. As established in Proposition 3, an unconstrained floor allows the policymaker to prevent credit rationing through the spread. When the floor binds, lowering the market rate stimulates

⁴⁹These signs compare local approximations at different reference points. The coefficients remain fixed within each approximation; a single linear approximation around the efficient steady state does not capture the reversal across regimes.

⁵⁰Further details in Appendix B.

demand but also strengthens the incentive to ration credit. Loan-supply constraints may arise in either case.

Proposition 4 (Full current lending) *If all projects are funded in period t ($\ln(x_t) = 0$) and $r_t > \underline{r}_t$, the discretionary targeting rule is:*

$$\pi_t = -\frac{\lambda_x}{\kappa} (\tilde{y}_t - \alpha \ln(x_{t-1})) \quad (5.13)$$

which reduces to the standard rule $\pi_t = -(\lambda_x/\kappa)\tilde{y}_t$ when $\ln(x_{t-1}) = 0$.

When past lending was constrained ($\ln(x_{t-1}) < 0$), the credit wedge enters the targeting rule as an endogenous cost-push shock through the marginal cost channel (5.12). The central bank must tolerate a larger negative output gap for a given inflation rate. A binding ELB on the reserve rate does not by itself change this targeting rule: liquidity operations can still lower the market rate while the reserve rate remains fixed. If easing induces credit rationing, the targeting rule changes as described below; if the market rate reaches the reserve floor, further rate reductions are infeasible and the equality may not be attainable.

Proposition 5 (Credit rationing at the ELB) *When the ELB on the floor binds and credit is rationed ($\ln(x_t) < 0$), the discretionary targeting rule is:*

$$\pi_t = -\frac{\lambda_x}{\kappa} (\tilde{y}_t - \alpha \ln(x_{t-1})) + \frac{\Lambda_r}{\kappa} \xi_{r,t} \quad (5.14)$$

*where $\xi_{r,t} > 0$ is the shadow value of relaxing the credit-rationing constraint, satisfying:*⁵¹

$$\xi_{r,t} = \beta \lambda_x \frac{1 - \alpha}{1 + \varphi} \alpha + \beta \alpha \mathbb{E}_t [\Lambda_r \xi_{r,t+1} + \Upsilon_r \xi_{l,t+1}] \quad (5.15)$$

The final term in (5.14) is an inflation bias whose sign is governed by Λ_r . Since Proposition 5 is conditioned on active credit rationing, and $\Lambda_r < 0$ in this region, the bias is negative: the policymaker restrains liquidity provision, accepting lower inflation and a wider output gap to limit the degree of credit rationing.

The multiplier $\xi_{r,t}$ captures the discounted marginal welfare gain from relaxing the credit-rationing constraint. The first term in (5.15) is the direct productivity benefit of additional lending; the remaining terms capture the influence of expected future lending conditions. Expected future rationing ($\xi_{r,t+1} > 0$) generates a deflation bias, lowering expected inflation and increasing the need for current accommodation. Since $\Lambda_r < 0$, this reduces $\xi_{r,t}$ and attenuates the current deflation bias. Expected future loan-supply constraints ($\xi_{l,t+1} > 0$) work in the opposite direction: accommodation to support future lending raises expected inflation, reducing the need for current easing. Since $\Upsilon_r > 0$, this increases $\xi_{r,t}$ and strengthens the current deflation bias.

The strength of the credit-rationing channel also depends on the degree of adverse selection.

⁵¹Equation (5.15) holds under active rationing. The general condition, which holds in all regimes, is $\xi_{r,t} + \xi_{l,t} + \xi_{x,t} = \beta \lambda_x \frac{1 - \alpha}{1 + \varphi} \alpha + \beta \alpha \mathbb{E}_t [\Lambda_r \xi_{r,t+1} + \Upsilon_r \xi_{l,t+1}]$, where $\xi_{l,t} \geq 0$ and $\xi_{x,t} \geq 0$ are the multipliers on the liquidity and feasibility constraints respectively.

From (5.6), higher borrower risk or a larger share of risky firms raises $|\Lambda_r|$ in the rationing region, sharpening the trade-off between demand stimulus and lending incentives. This is consistent with Figure 4, where liquidity policies can become contractionary under high borrower risk.

5.5 Optimal policy under commitment

Under commitment from a timeless perspective, the policymaker accounts for how promised future policy affects current lending incentives through $\mathbb{E}_t[\tilde{y}_{t+1}]$ and $\mathbb{E}_t[\pi_{t+1}]$ in the rationing condition (5.5). Current policy also honours past commitments, introducing history dependence into the targeting rule.

Proposition 6 (Commitment targeting rule) *The commitment targeting rule is:*

$$\pi_t + \frac{\lambda_x}{\kappa}(\tilde{y}_t - \alpha \ln(x_{t-1})) = \mu_{\pi,t-1} + \frac{\Lambda_r}{\kappa}\xi_{r,t} + \frac{\Upsilon_r}{\kappa}\xi_{l,t} - \Omega_r\xi_{r,t-1} - \Omega_l\xi_{l,t-1} \quad (5.16)$$

where Ω_r, Ω_l are composite coefficients, and $\xi_{r,t}, \xi_{l,t}$ are the shadow values of the rationing and loan-supply constraints. Under credit rationing ($\xi_{r,t} > 0, \xi_{l,t} = 0$), the shadow value satisfies:⁵²

$$\xi_{r,t} = \frac{1}{\Gamma_r} \left[\beta\lambda_x \frac{1-\alpha}{1+\varphi} \alpha + \beta\alpha \mathbb{E}_t[\Lambda_r\xi_{r,t+1} + \Upsilon_r\xi_{l,t+1}] \right] \quad (5.17)$$

When $\ln(x_t) = 0$ and $\ln(x_{t-1}) = \xi_{r,t-1} = \xi_{l,t-1} = 0$, the rule reduces to the standard price-level targeting rule $\pi_t = -(\lambda_x/\kappa)(\tilde{y}_t - \tilde{y}_{t-1})$.

The commitment targeting rule (5.16) differs from the discretionary rule (5.14) in two respects. First, the lagged multipliers $\mu_{\pi,t-1}$, $\xi_{r,t-1}$, and $\xi_{l,t-1}$ introduce history dependence. The lagged NKPC multiplier $\mu_{\pi,t-1}$ generates the standard price-level targeting property. The terms in $\xi_{r,t-1}$ and $\xi_{l,t-1}$ propagate past lending constraints into current policy. Since $\Omega_r < 0$ under rationing,⁵³ past rationing ($\xi_{r,t-1} > 0$) raises inflation through $-\Omega_r\xi_{r,t-1} > 0$: the policymaker delivers accommodation that was committed to during the rationing episode in order to raise inflation expectations and ease the credit constraint at that time. It is the anticipation of this accommodation that reduces the severity of rationing under commitment relative to discretion.

Second, commitment changes the shadow value of relaxing the credit-rationing constraint. Current rationing calls for future accommodation, which the policymaker must subsequently deliver. Avoiding rationing therefore reduces both the direct productivity loss and the need for this future policy response. This is reflected in the factor $1/\Gamma_r > 1$ in (5.17), which increases $\xi_{r,t}$ for given expected future shadow values and strengthens the incentive to preserve lending margins today. At the same time, credible promises of future accommodation raise expected lending profitability and ease the current credit constraint. Commitment thus combines a stronger incentive to limit rationing with an additional means of doing so through expectations.

Corollary 3 *The optimal degree of credit rationing under commitment is less severe than under*

⁵²Under discretion, $\Gamma_r = 1$. Under commitment, $\Gamma_r = 1 - \alpha(\Lambda_y - \Lambda_r) < 1$ since $\Lambda_y > \Lambda_r$. See the proof of Proposition 6.

⁵³From the definition $\Omega_r \equiv [(1 + \kappa)\Lambda_r - \Lambda_y - \kappa\Lambda_\pi]/(\beta\kappa)$. Since $\Lambda_r \leq 0$, $\Lambda_y > 0$, and $\Lambda_\pi > 0$, $\Omega_r < 0$. See the proof of Proposition 6.

discretion: $x_t^{\text{commit}} > x_t^{\text{discr}}$.

This provides a new channel through which commitment matters at the ELB, beyond the standard real-rate channel. Forward guidance not only reduces real interest rates; it also directly eases credit conditions by raising the expected return on lending relative to the floor, weakening the credit-rationing friction.

The history dependence of the commitment solution also implies gradualism in the withdrawal of accommodation. The lagged multiplier $\xi_{r,t-1}$ carries past rationing episodes into the targeting rule, sustaining accommodation even after full lending is restored. This accommodation fulfils the promises that raised expected lending profitability and eased credit conditions during the rationing episode. In the context of unconventional policy, this provides a rationale for gradual quantitative tightening: the return to full lending does not remove the obligation to deliver previously promised accommodation.

6 Concluding remarks

This paper makes three broad contributions. First, it provides empirical evidence consistent with a link between interest-rate compression and bank lending incentives. Using euro area bank-level and aggregate data, the analysis documents that narrower spreads between market rates and the policy floor are associated with lower bank lending, consistent with the credit-rationing channel developed in the theoretical model.

Second, the paper develops a structural model that rationalises these findings. By embedding adverse selection into a New Keynesian framework with bank balance sheets and a policy corridor, the model delivers endogenous lending regimes and state-dependent transmission of liquidity policies. Liquidity injections operate through two opposing channels: a conventional demand channel and a credit-rationing channel that works through banks' lending incentives. Which channel dominates depends on risk, entry conditions, and the opportunity cost of lending, rather than on banks' access to funding. As a result, credit scarcity can arise for the real economy even as bank liquidity increases.

Third, the paper draws implications for monetary policy design. Endogenous lending distortions enter the welfare loss function as a credit wedge, generating a trade-off between stabilising inflation and closing the efficient output gap that is absent under full lending. Under discretion, optimal policy at the effective lower bound features a deflation bias: the policymaker restrains liquidity provision to preserve bank lending margins, with the bias increasing in the severity of adverse selection. Under commitment, forward guidance eases credit conditions by credibly promising future accommodation that raises expected lending profitability. This generates history dependence in optimal policy: past rationing episodes oblige future accommodation, providing a rationale for gradualism in quantitative tightening. The return to full lending therefore does not remove the obligation to deliver previously promised accommodation.

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Online Appendices for “*Monetary Policy and the Credit Rationing Effects of Liquidity*”

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Appendix A Model equilibrium conditions

The competitive general equilibrium is determined by the following conditions:

$$1 = \mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} \right] R_t \quad (\text{A.1})$$

$$\frac{W_t}{P_t} = - \frac{U'(H_t)}{U'(C_t)} \quad (\text{A.2})$$

$$\frac{W_t}{P_t} = (1 - \alpha) MC_t z_t \left(\frac{K_{t-1}}{H_t} \right)^\alpha \quad (\text{A.3})$$

$$R_t^s = \frac{\alpha MC_t z_t \left(\frac{K_{t-1}}{H_t} \right)^{\alpha-1} + (1 - \delta) q_t^K}{q_{t-1}^K} \Pi_{t-1,t}, \quad (\text{A.4})$$

$$R_t^r = \omega_t^r R_t^s \quad (\text{A.5})$$

$$K_t = (1 - \delta) K_{t-1} + I_t \left[1 - \frac{\phi_K}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right] \quad (\text{A.6})$$

$$1 = q_t^K \left(\left[1 - \frac{\phi_K}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right] - \phi_K \left(\frac{I_t}{I_{t-1}} \right) \frac{I_t}{I_{t-1}} \right) + \mathbb{E}_t \left[\Lambda_{t,t+1} q_{t+1}^k \phi_K \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2 \right]. \quad (\text{A.7})$$

$$Y_t^W = z_t K_{t-1}^\alpha H_t^{1-\alpha} \quad (\text{A.8})$$

$$Y_t = \frac{Y_t^W}{\Delta_t} \quad (\text{A.9})$$

$$Y_t = C_t + I_t \quad (\text{A.10})$$

$$1 = \xi \Pi_{t-1,t}^{\zeta-1} + (1 - \xi) \left(\frac{\Omega_{1,t}}{\Omega_{2,t}} \right)^{1-\zeta} \quad (\text{A.11})$$

$$\Delta_t = \xi \Pi_{t-1,t}^\zeta \Delta_{t-1} + (1 - \xi) \left(\frac{\Omega_{1,t}}{\Omega_{2,t}} \right)^{-\zeta} \quad (\text{A.12})$$

$$\Omega_{1,t} = Y_t \frac{1}{1 - \frac{1}{\zeta}} MC_t + \xi \beta \mathbb{E}_t \left[\Lambda_{t,t+1} \Pi_{t,t+1}^\zeta \Omega_{1,t+1} \right] \quad (\text{A.13})$$

$$\Omega_{2,t} = Y_t + \xi \beta \mathbb{E}_t \left[\Lambda_{t,t+1} \Pi_{t,t+1}^{\zeta-1} \Omega_{2,t+1} \right] \quad (\text{A.14})$$

$$R_t = \frac{\Pi^*}{\beta} \left(\frac{Y_t}{\bar{Y}} \right)^{\gamma_y} \left(\frac{\Pi_{t-1,t}}{\Pi^*} \right)^{\gamma_\pi} \quad (\text{A.15})$$

We then have the firm entry condition, and the capital market clearing linking the number of firms and capital:

$$F = \frac{\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (\eta + (1-\eta)(1-\lambda)(1-p_{t+1})x_t^s) R_{t+1}^s \right] - \eta}{1 - \mathbb{E}_t [\Lambda_{t,t+1} (\eta + (1-\eta)(\lambda x_t^s + (1-\lambda)x_t^r p_{t+1}))]} q_t^K \quad (\text{A.16})$$

$$K_t = (\eta + (1-\eta)(\lambda x_t^s + (1-\lambda)x_t^r)) f_t \quad (\text{A.17})$$

The bank entry condition for liquidity ϕ :

$$1 = \mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} \left(\underline{R}_t \left(1 - [\lambda x_t^s + (1-\lambda)x_t^r] \frac{1}{\phi_t} \right) + (\lambda x_t^s + (1-\lambda)(x_t^r - (1-p_{t+1})x_t^s)) \frac{1}{\phi_t} R_{t+1}^s \right) \right] \quad (\text{A.18})$$

and conditions for lending (x_t^r and x_t^s):

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (R_{t+1}^s - \underline{R}_t) \right] = \varrho_t - \psi_t \frac{1}{1-\lambda} + \varphi_t^r \frac{1}{1-\lambda} \quad (\text{A.19})$$

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} ((\lambda + (1-\lambda)p_{t+1}) R_{t+1}^s - \underline{R}_t) \right] = \varrho_t + \varphi_t^r - \varphi_t^s \quad (\text{A.20})$$

$$\varphi_t^s, \varphi_t^r, \varrho_t, \psi_t \geq 0 \quad (\text{A.21})$$

$$\varphi_t^s x_t^s = 0 \quad (\text{A.22})$$

$$\varphi_t^r (1 - x_t^r) = 0 \quad (\text{A.23})$$

$$\varrho_t (\bar{x}_t - \lambda x_t^s - (1-\lambda)x_t^r) = 0 \quad (\text{A.24})$$

$$\psi_t (x_t^r - x_t^s) \quad (\text{A.25})$$

$$\bar{x}_t = \min \{1, \phi_t\} \quad (\text{A.26})$$

For the baseline numerical simulations, we use the King, Plosser and Rebelo (1988, KPR) class of utility with external habits in consumption:

$$U(C_t, H_t) = \frac{\left([C_t - h\bar{C}_{t-1}]^{1-\chi} (1-H_t)^\chi \right)^{1-\sigma}}{1-\sigma} \quad (\text{A.27})$$

For numerical simulations, we use a standard New Keynesian calibration shown in Table 3. Parameters for the novel part of the model (share of corporates, share of risky smalls, risky firm default) come from the calibration in Swarbrick (2023).

Appendix B Simplified Model and Welfare

B.1 Setup and equilibrium conditions

The simplified version abstracts from capital accumulation and firm entry, fixing the number of firms at f . Firms require a single unit of outside finance, which converts into ω_t^i units of

Parameter	Description	Value
β	Household discount factor	0.992
σ	Coefficient of relative risk aversion	1
χ	Consumption–leisure weight	$\bar{H} = 1/3$
α	Capital share in production	0.30
δ	Capital depreciation rate (quarterly)	0.023
ξ	Calvo price stickiness	0.75
ζ	Elasticity of substitution across goods	7
γ_π	Taylor rule inflation coefficient	1.5
γ_y	Taylor rule output gap coefficient	0.25
π^*	Inflation target (quarterly)	0.005
$\underline{R}$	Policy floor relative to policy rate (annualised)	−50 bps
η	Share of corporate firms	0.50
λ	Share of safe small firms	0.775
F	Firm entry cost	0.15
$1 - \bar{p}$	Steady-state risky firm default rate (annual)	6%
ϕ_K	Investment adjustment cost parameter	0
h	Habits in consumption	0.5

Table 3: Baseline parameter values

productive capacity. We set $\pi^* = \eta = 0$ and adopt separable preferences:

$$U(C_t, H_t) = \frac{C_t^{1-\sigma}}{1-\sigma} - \varphi_H \frac{H_t^{1+\varphi}}{1+\varphi} \quad (\text{B.1})$$

Production is $y_t^i = Z_t[\omega_t^i]^\alpha [h_t^i]^{1-\alpha}$, with flexible labour markets implying $R_t^r = \omega_t^r R_t^s$. Denoting aggregate productivity $A_t \equiv Z_t[f x_{t-1}]^\alpha$ where $x_t \equiv \lambda x_t^s + 1 - \lambda$, steady-state labour subsidies s_W remove the monopolistic competition distortion, and a lending subsidy s_L ensures $x = 1$:

$$s_L = \max \left\{ 0, \frac{\beta \lambda [1 - (1 - \lambda)(1 - p)] + 1 - \lambda}{1 - (1 - \lambda)(1 - p)} - 1 \right\} \quad (\text{B.2})$$

The non-linear equilibrium conditions are:

$$Y_t = \frac{1}{\Delta_t} A_t H_t^{1-\alpha} \quad (\text{B.3})$$

$$1 = \beta \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \frac{R_t}{\Pi_{t,t+1}} \right] \quad (\text{B.4})$$

$$\frac{W_t}{P_t} = \varphi_H C_t^\sigma H_t^\varphi \quad (\text{B.5})$$

$$C_t = Y_t \quad (\text{B.6})$$

$$\frac{W_t}{P_t} = (1 + s_W)(1 - \alpha) M C_t \frac{\Delta_t Y_t}{H_t} \quad (\text{B.7})$$

$$A_t = \exp(z_t) [f x_{t-1}]^\alpha \quad (\text{B.8})$$

$$R_t^s = \left[1 + \alpha(1 + s_W) M C_t \frac{\Delta_t Y_t}{f x_{t-1}} \right] \Pi_{t-1,t} \quad (\text{B.9})$$

$$1 = \beta \mathbb{E}_t \left[\left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \left(\underline{R}_t + (\lambda x_t^s (1 + s_L) (R_{t+1}^s - \underline{R}_t) \right. \right. \\ \left. \left. + (1 - \lambda) ((1 + s_L) R_{t+1}^s - (1 - p_{t+1}^r) (1 + s_L) R_{t+1}^s x_t^s - \underline{R}_t) \right) \frac{1}{\phi_t} \right] \frac{1}{\Pi_{t,t+1}} \quad (\text{B.10})$$

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} [(\lambda - (1 - \lambda) (1 - p_{t+1}^r)) (1 + s_L) R_{t+1}^s - \lambda \underline{R}_t] \right] - \varrho_t \lambda - \psi_t = 0 \quad (\text{B.11})$$

$$\min \{ \varrho_t, (\phi_t - \lambda x_t^s - (1 - \lambda)) \} = 0 \quad (\text{B.12})$$

$$\min \{ \psi_t, (1 - x_t^s) \} = 0 \quad (\text{B.13})$$

together with the standard Calvo pricing conditions and the Taylor rule from Appendix A.

B.2 Key coefficients of the linearised credit wedge

Taking a log-linear approximation around the efficient steady state ($x = 1, \sigma = 1$) and expressing the credit wedge in terms of $(\tilde{y}_t, \pi_t, \hat{x}_t)$, the rationing condition can be written as:

$$\ln(x_t) = \Lambda_0 - \Lambda^D (r_t - \rho) + \Lambda^\nu \hat{\nu}_t + \Lambda_y \mathbb{E}_t[\tilde{y}_{t+1}] + \Lambda_\pi \mathbb{E}_t[\pi_{t+1}] + u_{r,t} \quad (\text{B.14})$$

where the coefficients governing the two channels of monetary policy on lending are:

$$\Lambda^D \equiv \frac{1 - \alpha}{1 + \alpha\varphi} \frac{R^s}{R^s - 1} \quad (\text{B.15})$$

$$\Lambda^\nu \equiv \frac{1 - \alpha}{1 + \alpha\varphi} \frac{\lambda R}{(1 + s_L)(R^s - 1)[\lambda - (1 - p^r)(1 - \lambda)]} \quad (\text{B.16})$$

$$\Lambda_y \equiv \frac{(1 - \alpha) + (1 + \varphi)}{1 + \alpha\varphi} \quad (\text{B.17})$$

$$\Lambda_\pi \equiv \frac{1 - \alpha}{1 + \alpha\varphi} \frac{R^s}{R^s - 1} = \Lambda^D \quad (\text{B.18})$$

$\Lambda^D > 0$ governs the demand channel: lower r_t raises aggregate demand and the expected return on lending. $\Lambda^\nu > 0$ governs the credit-rationing channel: a wider spread to the floor increases the opportunity cost of holding reserves relative to lending. When the floor is constrained at the ELB, $\hat{\nu}_t = r_t - \rho$ and lending depends on r_t through $\Lambda_r \equiv \Lambda^D - \Lambda^\nu$. The exogenous shock term is:

$$u_{r,t} \equiv \frac{1 - \alpha}{1 + \alpha\varphi} \mathbb{E}_t[z_{t+1}] + \frac{R^s}{R^s - 1} \frac{1 - \alpha}{1 + \alpha\varphi} \frac{p^r (1 - \lambda)}{[\lambda - (1 - p^r)(1 - \lambda)]} \mathbb{E}_t[\hat{p}_{t+1}^r] \quad (\text{B.19})$$

The loan-supply constraint, when binding, yields an analogous condition with composite Υ -coefficients.

B.3 Welfare loss function

Taking a second-order expansion of utility around the efficient steady state with $\sigma = 1$, using $\hat{c}_t = \hat{y}_t$ and $\hat{y}_t - \hat{a}_t = \tilde{y}_t - \alpha \ln(x_{t-1})$:

$$\frac{U(C_t, H_t) - U(C, H)}{U_C Y} = -(\hat{\delta}_t - \hat{a}_t) - \frac{1}{2} \frac{1+\varphi}{1-\alpha} (\tilde{y}_t - \alpha \ln(x_{t-1}))^2 + \text{t.i.p.} \quad (\text{B.20})$$

The quadratic term is the natural (flex-price) output gap $\tilde{y}_t - \alpha \ln(x_{t-1})$, which equals zero when prices are flexible: from the marginal cost expression $\widehat{mc}_t = \frac{1+\varphi}{1-\alpha} (\tilde{y}_t - \alpha \ln(x_{t-1}))$.

Using the standard Calvo result $\sum \beta^t \hat{\delta}_t = \frac{\zeta \xi}{2(1-\beta\xi)(1-\xi)} \sum \beta^t \pi_t^2$ and $\hat{a}_t = z_t + \alpha \ln(x_{t-1})$, defining the welfare loss $\mathcal{L} \equiv -\mathbb{E}_0 \sum \beta^t \widehat{U}_t$ and normalising:

$$\mathcal{L} = \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[-\lambda_x \frac{1-\alpha}{1+\varphi} \alpha \ln(x_{t-1}) + \frac{1}{2} (\pi_t^2 + \lambda_x (\tilde{y}_t - \alpha \ln(x_{t-1}))^2) \right] \quad (\text{B.21})$$

where $\lambda_x \equiv \frac{1+\varphi}{1-\alpha} \frac{(1-\beta\xi)(1-\xi)}{\zeta\xi} = \frac{\kappa}{\zeta}$. Without the credit friction ($\ln(x_{t-1}) = 0$), this collapses to the standard $\frac{1}{2} [\pi_t^2 + \lambda_x \tilde{y}_t^2]$.

Appendix C Derivations and proofs

C.1 Lending facility use

Proposition 7 *Banks never use the central bank standing lending facility if $\bar{R}_t > R_t^p$. However, banks will use the deposit (reserve) facility in equilibrium and so the opportunity cost of lending is $\underline{R}_t \forall t$.*

Proposition 7 establishes that without a source of liquidity risk, the banks will never resort to using the central bank lending facility if this charges a higher interest rate than the main liquidity operation. So it truly is a last resort facility.

Proof: Without a source of liquidity risk, banks know their liquidity requirements with certainty at the time of portfolio allocation. Since funds are available through the main refinancing operation at rate $R_t^p < \bar{R}_t$, a bank facing a liquidity shortfall can always obtain funds at R_t^p rather than borrowing at the marginal lending facility rate $\bar{R}_t$. The lending facility is therefore strictly dominated and is never used in equilibrium. $\square$

C.2 Proof of proposition 1

The bank's first-order conditions for x_t^r and x_t^s are:

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (R_{t+1}^s - \underline{R}_t) \right] = \varrho_t - \frac{\psi_t}{1-\lambda} + \frac{\varphi_t^r}{1-\lambda} \quad (\text{C.1})$$

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} ((\lambda + (1-\lambda)p_{t+1}) R_{t+1}^s - \underline{R}_t) \right] = \varrho_t + \varphi_t^r - \varphi_t^s \quad (\text{C.2})$$

Subtracting (C.1) from (C.2):

$$-(1 - \lambda)\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (1 - p_{t+1}) R_{t+1}^s \right] = -\varphi_t^s + \frac{\psi_t}{1 - \lambda} - \frac{\lambda}{1 - \lambda} \varphi_t^r \quad (\text{C.3})$$

The left-hand side is strictly negative since $(1 - p_{t+1}) > 0$, $R_{t+1}^s > 0$, and $\Lambda_{t,t+1}/\Pi_{t,t+1} > 0$. If $x_t^s > 0$ then $\varphi_t^s = 0$ by complementary slackness, giving:

$$\frac{\psi_t}{1 - \lambda} - \frac{\lambda}{1 - \lambda} \varphi_t^r < 0 \quad \Rightarrow \quad \psi_t < \lambda \varphi_t^r \quad (\text{C.4})$$

Since $\psi_t \geq 0$, this requires $\varphi_t^r > 0$, which by complementary slackness implies $x_t^r = 1$.

C.3 Proof of proposition 2

By Proposition 1, $x_t^r = 1$ whenever $x_t^s > 0$, so $\varphi_t^r > 0$ in all three regimes. Under the maintained assumption $\varphi_t^s = 0$, combining the bank's first-order conditions for x_t^r and x_t^s yields:

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (\lambda(R_{t+1}^s - \underline{R}_t) - (1 - \lambda)(1 - p_{t+1})R_{t+1}^s) \right] = \lambda \varrho_t + \psi_t \quad (\text{C.5})$$

Since $\varrho_t, \psi_t \geq 0$, the LHS is non-negative.

No credit rationing. When (4.1) holds strictly, the LHS of (C.5) is positive, requiring $\lambda \varrho_t + \psi_t > 0$: either the loan-supply constraint binds ($\varrho_t > 0$) or there is pooling ($\psi_t > 0$). In either case, rationing does not occur.

Regime 3 (High-risk separating). When (4.1) holds with equality, $\lambda \varrho_t + \psi_t = 0$, so $\varrho_t = \psi_t = 0$: there is no pooling and the loan-supply constraint is slack (banks hold excess reserves). Banks optimally set $x_t^s < 1$ because the marginal revenue from a safe loan, $\lambda(R_{t+1}^s - \underline{R}_t)$, exactly equals the marginal information rent cost, $(1 - \lambda)(1 - p_{t+1})R_{t+1}^s$.

Regimes 1 and 2. When no rationing occurs, the regime depends on whether equilibrium bank liquidity is sufficient to fund all firms. The bank free-entry condition determines ϕ_t . When $x_t^s = 1$ and $\phi_t \geq 1$, evaluating the free-entry condition yields condition (4.2).

Regime 1 (Pooling). When both (4.1) and (4.2) hold, $\phi_t \geq x_t = 1$ and $\psi_t > 0$ enforces pooling. All firms are funded with excess reserves ($\varrho_t = 0$).

Regime 2 (Low-risk separating). When (4.1) holds but (4.2) fails, the expected return on the bank's portfolio is insufficient to attract enough entry for $\phi_t \geq 1$. The loan-supply constraint binds ($\varrho_t > 0$, $\phi_t = x_t < 1$), restricting $x_t^s < 1$ despite positive lending incentives ($\psi_t = 0$). $\square$

C.4 Proof of proposition 3

From Proposition 2, credit rationing occurs when:

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (\lambda(R_{t+1}^s - \underline{R}_t) - (1 - \lambda)(1 - p_{t+1})R_{t+1}^s) \right] = 0 \quad (\text{C.6})$$

The interest on reserves $\underline{R}_t$ enters this condition with coefficient $-\lambda < 0$. For any given R_{t+1}^s and p_{t+1} , reducing $\underline{R}_t$ (equivalently, raising $\nu_t = r_t - \underline{r}_t$) strictly increases the LHS. Away from the effective lower bound, the policymaker can always lower $\underline{R}_t$ sufficiently to make the LHS strictly positive, ensuring condition (4.1) holds and credit rationing does not occur.

The loan-supply constraint threshold (4.2):

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} [1 - (1 - \lambda)(1 - p_{t+1})] R_{t+1}^s \right] \geq 1 \quad (\text{C.7})$$

does not depend on $\underline{R}_t$. The spread instrument therefore cannot prevent this constraint from binding.

The linearised model confirms this structure: the rationing condition (5.7) contains $\Lambda^\nu \hat{\nu}_t$ with $\Lambda^\nu > 0$, so widening the spread relaxes the constraint, while the liquidity condition (5.8) is independent of $\hat{\nu}_t$. $\square$

C.5 Proof of proposition 4

When $\ln(x_t) = 0$, the feasibility constraint binds ($\xi_{x,t} \geq 0$) while the rationing and loan-supply constraints are slack ($\xi_{r,t} = \xi_{l,t} = 0$). From the FOC for r_t : $\mu_{y,t} = -\Lambda_r \xi_{r,t} - \Upsilon_r \xi_{l,t} = 0$. From the FOC for π_t : $\mu_{\pi,t} = -\pi_t$. Substituting both into the FOC for $\tilde{y}_t$:

$$\lambda_x(\tilde{y}_t - \alpha \ln(x_{t-1})) + \kappa \pi_t = 0 \quad (\text{C.8})$$

which gives (5.13). When $\ln(x_{t-1}) = 0$, this reduces to $\pi_t = -(\lambda_x/\kappa)\tilde{y}_t$. $\square$

C.6 Proof of proposition 5

When the ELB binds, $\nu_t = r_t - \underline{r}^{ELB}$ and the rationing constraint depends on r_t through $-\Lambda_r(r_t - \rho)$. Under active rationing ($\xi_{r,t} > 0$) with the loan-supply constraint slack ($\xi_{l,t} = 0$) and $\ln(x_t) < 0$ ($\xi_{x,t} = 0$), the discretionary FOCs give: $\mu_{\pi,t} = -\pi_t$; $\lambda_x(\tilde{y}_t - \alpha \ln(x_{t-1})) + \kappa \pi_t + \mu_{y,t} = 0$; and $\mu_{y,t} = -\Lambda_r \xi_{r,t}$. Combining yields (5.14).

For the recursion, since $\ln(x_t)$ affects welfare only from $t + 1$ onwards, the FOC for $\ln(x_t)$ is:

$$\xi_{r,t} = \beta \lambda_x \frac{1 - \alpha}{1 + \varphi} \alpha + \beta \lambda_x \alpha \mathbb{E}_t[\tilde{y}_{t+1} - \alpha \ln(x_t)] + \beta \kappa \alpha \mathbb{E}_t[\pi_{t+1}]$$

The targeting rule holds in all future states, giving $\lambda_x(\tilde{y}_{t+1} - \alpha \ln(x_t)) + \kappa \pi_{t+1} = \Lambda_r \xi_{r,t+1} + \Upsilon_r \xi_{l,t+1}$. Substituting yields (5.15). $\square$

C.7 Proof of proposition 6

Under commitment, π_t and $\tilde{y}_t$ appear in period- $(t-1)$ constraints through expectations, generating lagged multiplier terms absent under discretion. The FOCs are:

$$\pi_t + \mu_{\pi,t} - \mu_{\pi,t-1} - \frac{1}{\beta} \mu_{y,t-1} - \frac{\Lambda_\pi}{\beta} \xi_{r,t-1} - \frac{\Upsilon_\pi}{\beta} \xi_{l,t-1} = 0 \quad (\text{C.9})$$

$$\lambda_x(\tilde{y}_t - \alpha \ln(x_{t-1})) - \kappa \mu_{\pi,t} + \mu_{y,t} - \frac{1}{\beta} \mu_{y,t-1} - \frac{\Lambda_y}{\beta} \xi_{r,t-1} - \frac{\Upsilon_y}{\beta} \xi_{l,t-1} = 0 \quad (\text{C.10})$$

$$\mu_{y,t} + \Lambda_r \xi_{r,t} + \Upsilon_r \xi_{l,t} = 0 \quad (\text{C.11})$$

Multiplying (C.9) by κ , adding (C.10), and using (C.11) to eliminate $\mu_{y,t}$ and $\mu_{y,t-1}$ yields (5.16), where $\Omega_r \equiv [(1 + \kappa)\Lambda_r - \Lambda_y - \kappa\Lambda_\pi]/(\beta\kappa)$ and Ω_l is defined analogously.

For the recursion, the FOC for $\ln(x_t)$ is:

$$\xi_{r,t} + \xi_{l,t} + \xi_{x,t} = \beta\lambda_x \frac{1-\alpha}{1+\varphi} \alpha + \beta\lambda_x \alpha \mathbb{E}_t[\tilde{y}_{t+1} - \alpha \ln(x_t)] - \beta\kappa \alpha \mathbb{E}_t[\mu_{\pi,t+1}] \quad (\text{C.12})$$

Using the forward targeting rule and the forward (C.9) to express $\lambda_x(\tilde{y}_{t+1} - \alpha \ln(x_t)) - \kappa\mu_{\pi,t+1}$ in terms of multipliers, the $\mu_{\pi,t}$ and π_{t+1} terms cancel, leaving (under rationing, $\xi_{l,t} = \xi_{x,t} = 0$):

$$\Gamma_r \xi_{r,t} = \beta\lambda_x \frac{1-\alpha}{1+\varphi} \alpha + \beta\alpha \mathbb{E}_t[\Lambda_r \xi_{r,t+1} + \Upsilon_r \xi_{l,t+1}]$$

where $\Gamma_r = 1 - \alpha(\Lambda_y - \Lambda_r)$, giving (5.17). For the special case, setting $\xi_{r,t} = \xi_{l,t} = \xi_{r,t-1} = \xi_{l,t-1} = 0$ and $\ln(x_{t-1}) = 0$ in (5.16) gives $\pi_t + (\lambda_x/\kappa)\tilde{y}_t = \mu_{\pi,t-1}$. From (C.10) with zero credit multipliers: $\mu_{\pi,t} = (\lambda_x/\kappa)\tilde{y}_t$, so $\mu_{\pi,t-1} = (\lambda_x/\kappa)\tilde{y}_{t-1}$, yielding $\pi_t = -(\lambda_x/\kappa)(\tilde{y}_t - \tilde{y}_{t-1})$. $\square$

Appendix D Empirical Results

This appendix reports supplementary bank-level results supporting the aggregate analysis in Section 2.

D.1 Data and sample construction

Bank-level balance-sheet data are from S&P Global Market Intelligence (SNL Financial Fundamentals), covering all available banks operating in Austria, Belgium, France, Germany, Ireland, Italy, the Netherlands, Portugal, Slovenia and Spain over 2008Q1–2019Q4. The balance-sheet variables are cash and balances with central banks, $\ln Res_{i,t}$, and total gross loans, $\ln L_{i,t}$. Bank-level lending is normalised by the bank's average asset level over 2008Q1–2019Q4, and the same asset value is used as a sampling weight. All specifications include bank fixed effects; standard errors in Table 1 are clustered two ways, by bank and by quarter, and in Table 2 by bank.

The interbank market rate R_t is Eonia. Since October 2019 the ECB has targeted the euro short-term rate, replacing Eonia as the reference interbank rate (see Nicoloso and Tsonchev, 2019, Box 1, p.23). The interest on reserves $\underline{R}_t$ is the rate on the ECB's standing deposit facility. The spread to the floor, $s_t \equiv R_t - \underline{R}_t$, is measured per 25 basis points.

The controls $\mathbf{z}_t$ are total euro-area log real GDP, CPI (HICP) inflation, Gilchrist and Mojon's (2018) measure of credit risk of euro-area non-financial corporations, and the stock of total asset purchase programmes held by the Eurosystem for monetary policy purposes.

The bank characteristics $Z_{k,i}$ in equation (2.2) are bank-constant: each is bank i 's average ratio over 2008–2010, less the cross-bank mean and divided by the cross-bank standard deviation. Reserve-richness uses cash and balances at the central bank over total assets, deposit funding uses customer deposits over total assets, and capitalisation uses equity over total assets, all from

D.2 Robustness of the aggregate balance-sheet relationships

Table 4 reports the spread coefficient under three estimators. Because the spread is an aggregate, persistent series, two-way clustering may understate its standard error; the asset-weighted quarterly aggregate with a Newey-West correction addresses this, giving a loan coefficient of +0.041 ($t = 4.3$).

Significance rests on the weighting. The loan coefficient is significant at one percent under asset weighting and only marginally so under equal weighting (+0.034, $p = 0.09$), with the same sign and magnitude either way. This is expected: the effect concentrates in the largest banks, which hold almost all of the assets in the sample (Table 2), so asset weighting recovers the aggregate-representative effect.

The interbank-rate coefficient is negative and significant throughout. The implied effect of the interbank rate holding the floor fixed, the sum of the two coefficients, is +0.029 ($p = 0.03$) under asset weighting and a smaller, less precise +0.014 under equal weighting.

	$\ln L_t$	$\ln Res_t$
Asset-weighted, two-way cluster	0.048*** (0.013)	-0.172*** (0.062)
Equal-weighted, two-way cluster	0.034* (0.020)	-0.104* (0.057)
Asset-weighted aggregate, Newey-West	0.041*** (0.010)	-0.201 (0.131)

Notes: Each cell is the coefficient on the spread to the floor, s_t , per 25 basis points, from a separate regression with the controls of Table 1. The first two rows are the bank-level panel with bank fixed effects (4,161 loan and 5,762 reserve observations), differing only in whether observations are weighted by bank assets; standard errors are clustered by bank and quarter. The third row collapses the panel to 48 asset-weighted quarterly aggregates and applies a Newey-West correction at lag four. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 4: The floor-spread coefficient under alternative weighting and serial-correlation corrections

D.3 Cross-sectional heterogeneity across size cuts and fixed-effects schemes

Table 5 reports the reserve-richness interaction across size cuts, under both fixed-effects schemes of Table 2; the top-quartile row reproduces that table. The interaction is positive throughout the size distribution. Under bank and quarter fixed effects it is significant from the top 30% of banks through the top 20%, and insignificant only when the cut is broadened to the top half. Under core/periphery-by-quarter fixed effects it is significant at every cut and larger as the cut narrows. The deposit-funding and capitalisation interactions are insignificant at every cut and are omitted.

	Bank + quarter FE (1)	Core/periph. × quarter FE (2)	Banks	Obs.
Top 50%	0.006 (0.005)	0.010** (0.005)	77	2,076
Top 30%	0.015** (0.007)	0.019** (0.008)	70	1,932
Top 25% (quartile)	0.018** (0.007)	0.022*** (0.008)	62	1,784
Top 20%	0.014** (0.007)	0.024** (0.011)	53	1,562

Notes: Each cell is the coefficient on $s_t \times$ reserve-richness from the three-interaction specification of equation (2.2), estimated on banks above the stated percentile of mean assets. Standard errors clustered by bank in parentheses. Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 5: The reserve-richness interaction across size cuts

Appendix E Collateral extension

This appendix extends the bank's problem to allow firms to pledge personal wealth as collateral. We show that collateral shifts the threshold at which credit rationing occurs but does not eliminate the mechanism.

E.1 Setup

Each firm is endowed with personal wealth $\bar{c} \geq 0$ per unit of capital, independent of type. This wealth is unrelated to the project being financed and represents pledgeable assets such as the entrepreneur's personal real estate or savings. The bank's contract for type $i \in \{s, r\}$ is now (τ_t^i, x_t^i, c_t^i) : a repayment rate, an approval probability, and a collateral requirement $c_t^i \in [0, \bar{c}]$. If the firm repays, the collateral is returned. If the firm defaults, the bank seizes c_t^i and recovers its full value.¹

E.2 Modified constraints

Since safe firms never default, their collateral is always returned. Collateral is therefore costless for the safe type, and the participation constraint is unchanged from (3.2). This asymmetry is the key to the Bester mechanism: a risky firm mimicking the safe contract now faces expected collateral loss $(1 - p_{t+1})c_t^s$ per approved loan upon default. Higher c_t^s makes mimicking more costly, so the bank sets $c_t^s = \bar{c}$. With the participation constraint binding ($\tau_t^s = R_{t+1}^s$), the incentive compatibility constraint (3.3) becomes:

$$[p_{t+1}(R_{t+1}^r - \tau_t^r) - (1 - p_{t+1})c_t^r] x_t^r \geq [(1 - p_{t+1})(R_{t+1}^s - \bar{c})] x_t^s, \quad (\text{E.1})$$

where the right-hand side uses $p_{t+1}R_{t+1}^r = R_{t+1}^s$. The modified information rent per safe loan is $(1 - p_{t+1})(R_{t+1}^s - \bar{c})$, reduced by $(1 - p_{t+1})\bar{c}$ relative to the baseline.

¹Assuming no deadweight loss from seizure is conservative: it maximises the screening power of collateral and therefore provides the strongest test of whether collateral eliminates credit rationing.

E.3 Bank optimisation

The bank chooses $(x_t^s, x_t^r, \tau_t^r, c_t^r)$ to maximise expected lending profit:

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (\lambda x_t^s (R_{t+1}^s - \underline{R}_t) + (1 - \lambda) x_t^r [p_{t+1} \tau_t^r + (1 - p_{t+1}) c_t^r - \underline{R}_t]) \right], \quad (\text{E.2})$$

subject to the IC (E.1), feasibility $x_t^s \in [0, 1]$, $x_t^r \in [0, 1]$, and $c_t^r \in [0, \bar{c}]$.

The first-order conditions yield three results that parallel the baseline.

First, c_t^r is indeterminate: along the binding IC, any increase in c_t^r is exactly offset by a decrease in τ_t^r , leaving bank revenue and the equilibrium allocation unchanged. We set $c_t^r = 0$ without loss of generality.

Second, $x_t^r = 1$. Each additional risky loan earns expected net revenue $\mathbb{E}_t[\Lambda_{t,t+1} \Pi_{t,t+1}^{-1} (R_{t+1}^s - \underline{R}_t)] > 0$ and relaxes the IC by increasing the risky type's surplus from truthful revelation. The introduction of collateral does not alter this result.

Third, the first-order condition for x_t^s equates the marginal revenue from an additional safe loan with the marginal information-rent cost. Setting $c_t^r = 0$ and $x_t^r = 1$, interior rationing ($x_t^s < 1$) occurs when:

$$\mathbb{E}_t \left[\frac{\Lambda_{t,t+1}}{\Pi_{t,t+1}} (\lambda (R_{t+1}^s - \underline{R}_t) - (1 - \lambda)(1 - p_{t+1})(R_{t+1}^s - \bar{c})) \right] = 0. \quad (\text{E.3})$$

This generalises (4.4): the information-rent cost $(1 - \lambda)(1 - p_{t+1})R_{t+1}^s$ is replaced by $(1 - \lambda)(1 - p_{t+1})(R_{t+1}^s - \bar{c})$. The baseline is recovered by setting $\bar{c} = 0$.

E.4 Implications

Three results follow from (E.3).

1. *Threshold shift.* Credit rationing occurs when $\lambda(R_{t+1}^s - \underline{R}_t) < (1 - \lambda)(1 - p_{t+1})(R_{t+1}^s - \bar{c})$. Since the right-hand side is strictly decreasing in $\bar{c}$, higher collateral shrinks the rationing region. However, for any $\bar{c} < R_{t+1}^s$, the right-hand side remains strictly positive and rationing can occur for sufficiently high default risk or a sufficiently tight policy corridor. The entire structure of the model, including the dependence of rationing on $\underline{R}_t$ and the competing channels of liquidity policy, is preserved.
2. *Full separation.* The information rent is non-positive (and hence rationing is eliminated for all parameter values) if and only if $\bar{c} \geq R_{t+1}^s$. This requires the entrepreneur's pledgeable personal wealth per unit of capital to exceed the gross return on capital. Since the model's borrower population consists of firms requiring external finance precisely because they lack sufficient internal funds, this condition is not satisfied for the relevant margin.
3. *Procyclicality.* If $\bar{c}$ depends on asset prices (personal real estate, existing business equipment), it is procyclical. In downturns, p_{t+1} falls and $\bar{c}$ falls simultaneously, both pushing the information rent $(1 - p_{t+1})(R_{t+1}^s - \bar{c})$ higher and the rationing condition towards binding. The collateral screening mechanism is therefore weakest in the states of the world

where the credit-rationing channel is most relevant.

The above analysis considers the most favourable case for collateral as a screening device. The empirical picture is less clear-cut: much of the literature finds that *riskier* borrowers post more collateral, consistent with banks using collateral to mitigate losses on observably risky loans rather than to screen unobserved types (Berger and Udell, 1990; Jiménez et al., 2006). Even studies that find evidence for the screening channel find it operating alongside observed-risk explanations, not instead of them (Berger et al., 2016). This suggests that collateral in practice attenuates adverse selection but does not resolve it, reinforcing the case for the rationing mechanism studied in this paper.

The remaining equilibrium conditions (the loan-supply constraint threshold, the Phillips curve, the IS curve, and the welfare approximation) are unaffected, since collateral enters only through the modified information rent in the rationing condition. The corollaries to Proposition 3 continue to hold, with the sole addition that the rationing incentive now also falls in $\bar{c}$.

Appendix F Additional simulations and figures

This appendix contains the results from further simulations. Figure 7 repeats the high vs low risk economy simulations to a QE programme with investment adjustment costs.

We next consider preferences of the type proposed by Jaimovich and Rebelo (2009, JR), which allow for a weak short-run wealth effect on labour supply. These take the form:

$$U(C_t, H_t) = \frac{1}{1-\sigma} \left(C_t - \chi H_t^\theta X_t \right)^{1-\sigma} \quad (\text{F.1})$$

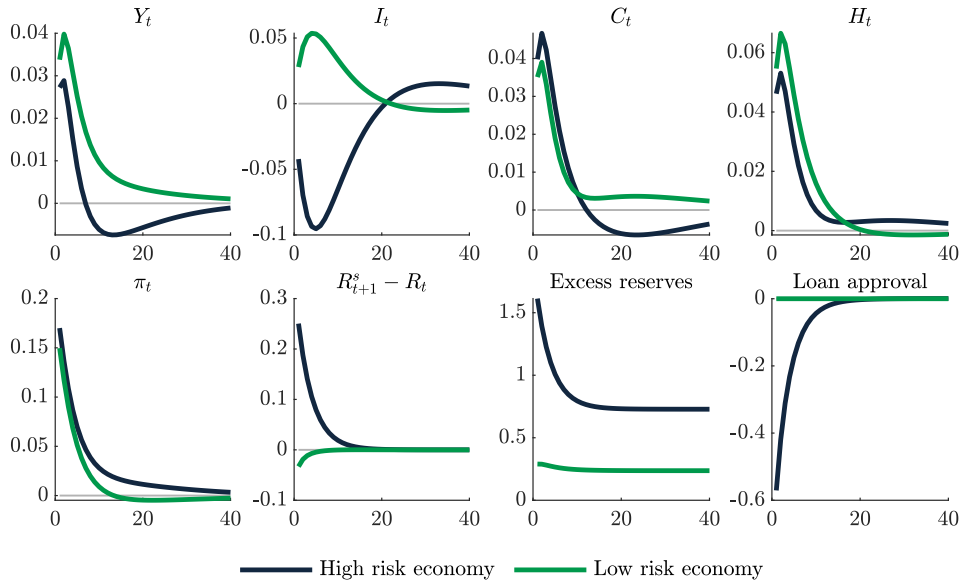


Figure 7: Impulse response functions to an unexpected temporary 'QE' programme in model with investment adjustment costs ($\phi_K = 4$). Plots in the top row show % deviation from steady state. The bottom row shows ppt deviation except for excess reserves which plots the ratio of reserves to stock of loans. Per annum risky firm default rate is 12% (high risk) and 2% (low risk).

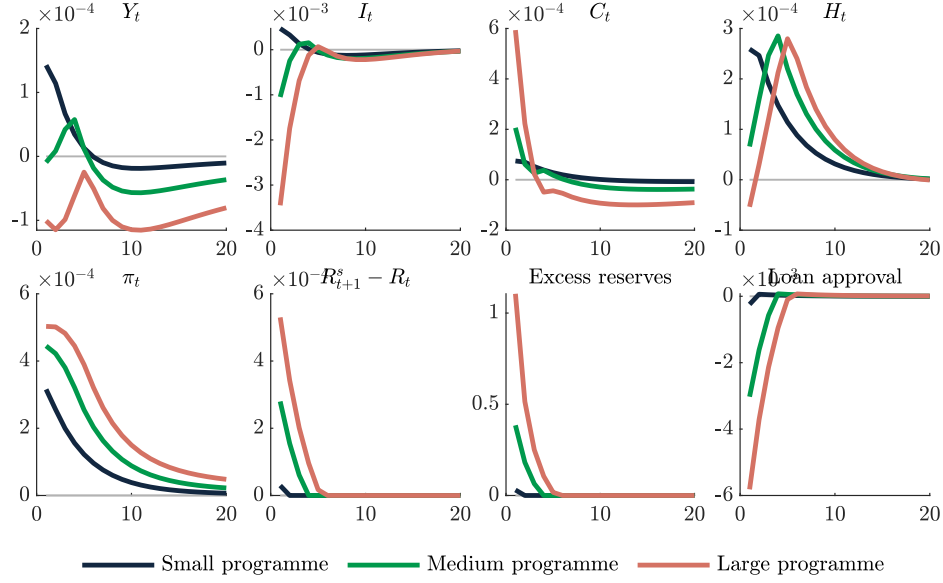


Figure 8: Impulse response functions to a temporary ‘QE’ programme with Jaimovich-Rebelo preferences, calibrated with a weak wealth effect on labour. Plots in the top row show % deviation from steady state. The bottom row shows ppt deviation except for excess reserves which plots the ratio of reserves to stock of loans.

where $X_t = C_t^\gamma X_{t-1}^{1-\gamma}$. When calibrated with ($\gamma = 0.2$) and an inverse Frisch elasticity $\theta = 2.5$, these preferences substantially dampen the consumption response to lower interest rates. Figures 8 and 9 shows the small/medium/large simulations with JR preferences without and with investment adjustment costs, respectively.

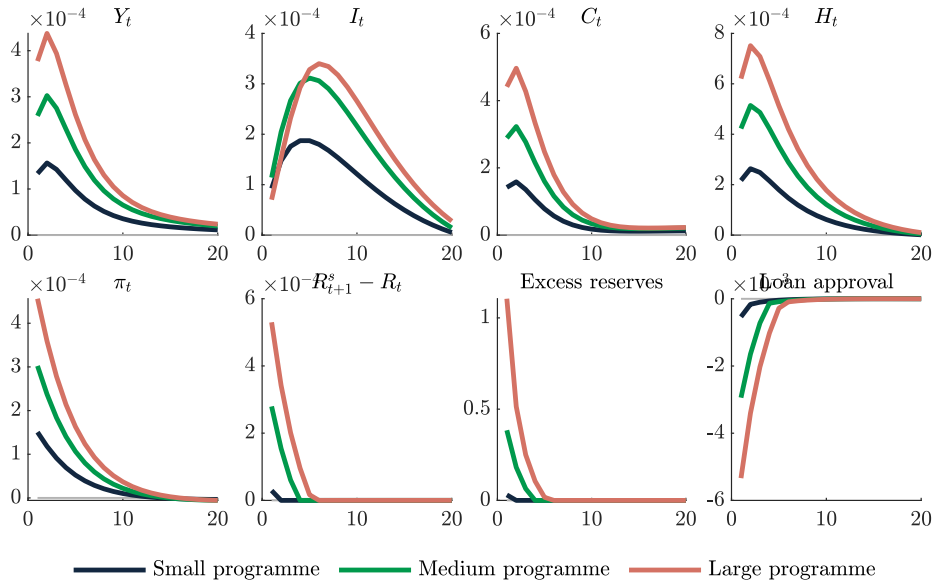


Figure 9: Impulse response functions to a temporary 'QE' programme with Jaimovich-Rebelo preferences, calibrated with a weak wealth effect on labour, and investment adjustment costs ($\phi_K = 4$). Plots in the top row show % deviation from steady state. The bottom row shows ppt deviation except for excess reserves which plots the ratio of reserves to stock of loans.